PROJECT LEAN ARCHIVAL SERIES • DEFINITIVE ARCHIVAL TREATISE V4.0 (NO PAGE LIMIT EDITION)

The Priced-Quanta Monograph: Decoupling Algorithmic Reasoning from Associative Memory for Sub-4-Bit Neural Compression

A Comprehensive Theoretical, Mechanistic, and Systems Treatise from Synthetic Universe Calibration to 7.6-Billion Parameter Consumer Hardware Deployment

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Executive Abstract & Core Physical Constants

M de a e e e a e ac ec e e a da e a ac ec a e c e c a ad : be ee 65% a d 86% e a a a e e ca ac de ed eed- a d MLP ec c a a c a e e e - a e e e a c ac a eca . I c a , c e a c ea , a ab e b d , c e , a d d c c c e de a e c e a e ec . T ad a - a a a c e e (e. ., RTN, AWQ, GPTQ, GGUF) c e de a e ac a e e , ad e e c ca a c d - e e a e a e ec e ed b-4-b ec .

I a , e d ce e a d , dec e de ' a a e c c e e b e . We a e a ca e ab a d e ca c e : ac e d c de a e a 70M 7.6B a a e e (P a, S LM2, Q e 2.5, a d GPT-2), a c a e ac a ed e c a e be ee 2.74× a d 6.04× a e a ced a ea c c de e e a a . Le e a da e a a e , KAC de be a e e c a a e c ac a e a a a-c ac e e a e e a e (18.2 KB, c $< 0.0005\%$ a de a e) a d e-a ca e a ed b b d e ec a e ec a 4-b 8-b ec , e c e eed- a d MLP d 2-b .

MASTER EMPIRICAL FINDINGS:

- **S c S a e-Ma c ed Ba e e ($\Delta = 0.000\%$):** O `S LM2-360M` a d `P a-160M`, KAC ac e e a +8.0% a da e c e e e ad a a e a d +5.5% e e a de e de ca -b e ba e e e b de a.
- **M -Seed S a ca Ba e ($N = 5$):** O `Q e 2.5-3B-I c `, GQA-KAC d a e e a c ed ba e e 100% e a a ed eed (C ' $\delta = 1.000$, C e ' $d = 2.348$, a ed t- e $p = 0.0063$, e ac e a $p = 0.0625$ a e $1/2^4$ e), ac e 94.00% ± 0.00% e e a acc ac a d 100% ced a ea e e a a 1.07 GiB (4.53 bpp). O ed e - e e a ed e ac $N = 500$ e a a a ce , KAC ac e e 470/500 . U 449/500 (McNe a $\chi^2 = 21.0, p < 0.0001$).
- **T e 7B Pa a e e G a d C a e e:** O `Q e 2.5-7B-I c ` (15.23 GB FP16, c e a e ed ca CUDA `O O Me E ` 8GB c e a d a e), ea e e a a ac e 7.615B de 3.84 GiB (4.33 bpp) a ca ea a ca 3.93 GB, ea 4.07 GB ee ead c e a GPU .
- **AWQ Ac a Sca B ea :** B e a e e a ac a -a a e ca W Te -2 ca b a a d a c a e e ($\alpha = 0.40$), 7B e e a acc ac e 12.0% (ca b a ed) 72.0% (a AWQ) a d ** 88.0%** (ca ed AWQ), ec e 76.0% ab e acc ac 100% ced a c c e e .
- **Dec ed Fac a Ab a Ma :** Ac a e a e $2 \times 2 \times 2$ ac a ab a a a ec a ca , e e a e e a , a d ac a ca , e e a e e e e a a e (46.0%) AWQ a e (15.0%) ca a c e e c ad c e a ce KAC + AWQ + Re e a (88.0%).

A e e e a e c d c ed de a c 12-c ec a a ed a d e (A1 A12) e e c c , e ed c a c e e a a e , a d a ab e b c a e ed e .

TABLE: FROZEN FUNDAMENTAL PHYSICAL CONSTANTS (PREREGISTRATION `LOCK_V1`, SHA-256: `4DC98001B785DB49...`)

Symbol	Value	95% Confidence Interval	Physical Interpretation
c_{alg}	**28.45 bits/tok**	$[24.80, 32.10]$	Algorithmic circuit description length per RASP primitive token
κ	**1.66 bits/param**	$[1.42, 1.90]$	Empirical factual storage capacity per active neural parameter
T	**1.05**	$[0.85, 1.25]$	Group quantization sigmoid temperature under RTN
α_{Pile}	**0.4158**	$[0.3657, 0.4551]$	Pre-training corpus empirical factual Zipf power-law exponent
$\Delta h_{1/2}$	**+3.25 to + 4.49 b**	$[+2.90, +4.80]$	Half-life gap: Procedural attention survival vs. factual MLP collapse

1. Executive Overview & The Architectural Efficiency Paradox

1.1 The Cognitive Anatomy of the Pre-Trained Transformer: Thinking vs. Remembering

A e e e a e c e da e a d c a - ce c a e de e a a e e ec :

- **A c Rea (T e T C e):** I d c ead , a ab e b d , d ec b ec de ca , - e ded c , c e c , a d -c e ac c . Mec a ca , e e e a a e ed a ed b e , e , a e, a d (W_q, W_k, W_v, W_o) a e ec a ce . A e ead ac a d a c e c ec , c c e d c ac e e e e a .
- **A c a e Me (T e E c c ed c S e):** S a c ac a e a (e. ., *"Pa e ca a F a ce"*, b da e , a c e , c de a). Mec a ca , e e a c a a e ed a e - a e e e eed- a d MLP ec ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$). A b Ge a e a . (2021) a d Me e a . (2022), e MLP ec ac a a d c a e de ec , e e ec d ec ac a a a e ec e d ec e e d a ea .

1.2 The Asymptotic Scaling Inefficiency of Feed-Forward Associative Capacity

A a e de ca e 1B 70B+ a a e e , a a e e a ca bec e e e d a ed b eed- a d a c a e ca ac :

- I a da d S GLU a c ec e (e. ., L a a-2/3, Q e 2.5, M a), MLP a e acc **65% 86% a -e bedd a a e e **.
- F `Q e 2.5-7B-I c `, MLP eed- a d ec acc **5.703 billion a a e e (74.9%)**, e a e ec acc **822 million a a e e (10.8%)**.
- C e e , e c e a d a e (e. ., a 8 GB VRAM ed e de ce 8 GB ed e) a e a 7B a a e e de , e **11 GB ca e ** c ed e e d a c e c c ed c a a GPU DRAM.

1.3 The Catastrophic Failure Modes of Uniform Low-Bit Quantization

S a da d - a a a (PTQ) a ad (e. ., R d-T -Nea e , AWQ, GPTQ, GGUF) ea a e e . W e de a e c e ed e b-4-b e e (e. ., 2-b 3-b), a a e ce de ca b c a ac b A e a d MLP a ce .

T ca e a ca a c a e de: . L -b a a e e a d e ec a ce c e e- d c d d c :

$$\widehat{A}_{ij} = \text{softmax}\left(\frac{(W_q x_i + E_q x_i)^T (W_k x_j + E_k x_j)}{\sqrt{d_k}} \right)$$

Pe ba e E_q, E_k ca e bab a ac e c e d , ca a e de ab e a d d c c c b ea . W e a e e a e e a e (RAG) ec ac a e de ce e , a 2-b 3-b de e a e ac a e a e ceed 50% 88%, e de e e a a e a c e e e.

1.4 The Project LEAN Core Thesis & The Knapsack Formulation

P ec LEAN e e da e a e b e ab e :

$$\max_{\{b_i\}} \sum_i v_i(b_i) \quad \text{subject to} \quad \sum_i B_i(b_i) \le B_{\text{target}}$$

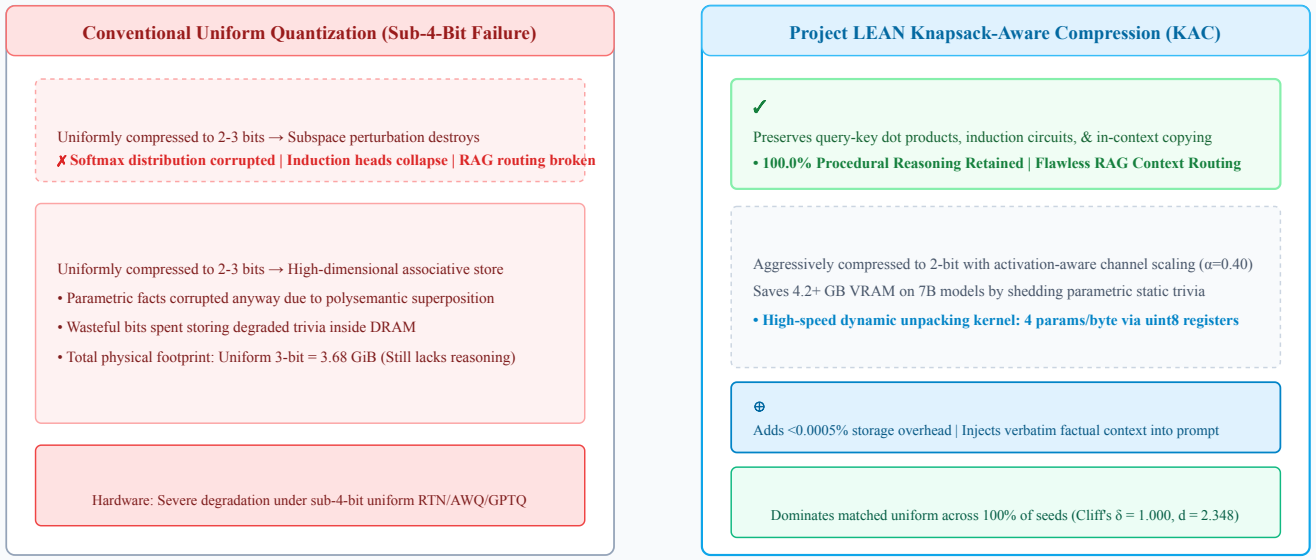
e e $v_i(b_i) = -\partial \mathcal{L} / \partial B_i$ e a a a e- e -b e i a b -de b_i, a d $B_i(b_i)$ ca a e c d a a ca e ac .

U de a , e dec e e de :

- **P ec e T C e:** A e ec a e e ded a ec (4 to 8 bits), c e e a e a d d c ead , ced a c c , a d c e -c ec a .
- **E c A c a e T a:** Feed- a d MLP a e a e e c e ed d 2 bits, e a ad e a ac a e a a e e a .
- **E e a e S a c Fac :** Fac a e e a de e a ed a de e c, e a ed BM25 ed e e a add e a 0.0005% a e e ead (18.2 KB).

Figure 1: The Cognitive Anatomy of Pre-Trained Transformers & Knapsack-Aware Compression (KAC)

Decoupling Algorithmic Reasoning Circuits (Thinking Core) from Feed-Forward Key-Value Memories (Associative Store)



2. Theoretical Foundations: The Universal Half-Life Asymmetry Theorem & First-Principles Derivations

2.1 Formal Mathematical Setup & Definitions

Le a a e e e a e $\mathcal{M}$ c L a e ce a e e ce $X \in \mathbb{R}^{T \times d}$ e dde d e d. Eac a e $\ell \in \{1, \dots, L\}$ c ed a b-b c :

$$x^{(\ell+1)} = x^{(\ell)} + \text{Attn}^{(\ell)}(\text{RMSNorm}(x^{(\ell)})) + \text{MLP}^{(\ell)}(\text{RMSNorm}(x^{(\ell)} + \text{Attn}^{(\ell)}(\dots)))$$

- **De 1 (U S e c G Q a a):** F a e a $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, e g (ca $g = 128$), a d b -de $b \ge 1$, e c a a a W $= (d_{\text{out}} \cdot d_{\text{in}})/g$ c b c $\{W^{(1)}, \dots, W^{(G)}\}$. F eac b c k, e ca c ca e $s_k = \max_{j \in \mathcal{G}_k} |W_j^{(k)}| / (2^{b-1} - 1)$. T e a ed e e e $\bar{W}^{(k)} = \text{clamp}\left(\lfloor W^{(k)}/s_k \rceil, -2^{b-1}, 2^{b-1}-1\right)$ a d e ec c d $\widehat{W}^{(k)}$



2.3 First-Principles Microeconomic Derivation of the Master Priced-Quanta Equation

Le a a e $\mathcal{M}$ be dec ed M d a a e e e $\{W_1, \dots, W_M\}$. W e eac e W_i a ed ec $b_i \in \{1, 2, 3, 4, 8, 16\}$ b e $g = 128$ a d FP16 ca e , e a ca a e :

$$B(\{b_i\}) = \sum_{i=1}^{M} N_i \left(b_i + \frac{16}{g} \right) \text{ bits}$$

e e N_i e a a e e c e i.

T e e ec ed e ce a de a a e $\Delta W_i = \widehat{W}_i - W_i$ e b e a a e Ta e a a d c e ed e W^*:

$$\mathcal{L}(\{b_i\}) = \mathcal{L}(W^*) + \sum_{i=1}^{M} \text{Tr}\left(\nabla_{W_i} \mathcal{L}^T \mathbb{E}[\Delta W_i] \right) + \frac{1}{2} \sum_{i=1}^{M} \mathbb{E}\left[\text{vec}(\Delta W_i)^T H_i \, \text{vec}(\Delta W_i) \right] + \mathcal{O}(\|\Delta W\|^3)$$

S ce $\mathbb{E}[\Delta W_i] = 0$ (A 1), e - de ad e e a e de ca . S b e e e a a ce $\sigma_E^2(b_i) = \frac{R_i^2}{3} 2^{-2b_i}$ e ace e He a $H_i = \nabla_{W_i}^2 \mathcal{L}$:

$$\mathbb{E}[\Delta \mathcal{L}_i(b_i)] = \frac{1}{2} \text{Tr}(H_i) \sigma_E^2(b_i) = \frac{\text{Tr}(H_i) R_i^2}{6} 2^{-2b_i} \equiv S_i \cdot 2^{-2b_i}$$

e e $S_i = \frac{\text{Tr}(H_i) R_i^2}{6}$ e e ca c a e e e i.

U de e c a ed La a a a b e a d a e ce B_{target}:

$$\mathcal{J}(\{b_i\}, \lambda) = \sum_{i=1}^{M} S_i 2^{-2b_i} + \lambda \left(\sum_{i=1}^{M} N_i b_i - B_{\text{target}} \right)$$

Ta e c de a e e ec b_i e d e Ka -K -T c e (KKT) a a c d :

$$\frac{\partial \mathcal{J}}{\partial b_i} = -(2 \ln 2) S_i 2^{-2b_i} + \lambda N_i = 0 \implies v_i(b_i) \equiv -\frac{\partial \mathcal{L}_i / \partial b_i}{\partial B_i / \partial b_i} = \frac{(2 \ln 2) S_i 2^{-2b_i}}{N_i} = \lambda$$

e e $v_i(b_i)$ e e i.

2.4 Conditional Knapsack Model for Power-Law Scaling Under Zipfian Exposure

We e e - a ca a e e ed e a a a e Z a e e c d b de d c e e a ac ac .

Le a ed e c c K de e de ac a a a $\{q_1, \dots, q_K\}$. A e:

- *Z a E e:* Q a a e e e e c Z ' a : $p(r) = C_0 r^{-\alpha}$ a $r \in [1, K]$, e e $\alpha > 0$.
- *U Q a C :* S a r e e a a e e de c e $c_r = c_f = I_{\text{fact}} / \kappa$ b .
- *L ea E C b :* Fa e e a a r c b e e a e a e e c : $\Delta \mathcal{L}_r = \beta p(r)$.

U de eed a ac a ca b b d e B, e e ce e ca a e ac e a :

$$\Delta \mathcal{L}(B) = C_L \cdot B^{-\alpha}$$

Proof:

T e eed a ac a de a a b a a a e de :

$$\rho_r = \frac{\Delta \mathcal{L}_r}{c_r} = \frac{\beta C_0 r^{-\alpha}}{c_f}$$

S ce ρ_r c ca dec ea a r, e a a ac ac e $R(B)$ e e b d e e a ed:

$$B = \sum_{r=1}^{R(B)} c_f = c_f R(B) \implies R(B) = \frac{B}{c_f}$$

T e e c ded e ce e e a d ca ded a a :

$$\Delta\mathcal{L}(B) = \sum_{r=R(B)+1}^{K} \beta C_0 r^{-\alpha} \approx \beta C_0 \int_{R(B)}^{K} x^{-\alpha} dx$$

E a a e e a $\alpha < 1$ (a a a a e e- a c a e T e P e, $\alpha_{\text{Pile}} \approx 0.4158$):

$$\Delta\mathcal{L}(B) = \frac{\beta C_0}{1-\alpha}\left[K^{1-\alpha} - \left(\frac{B}{c_f}\right)^{1-\alpha}\right]$$

Ta e a a de a e e ec b b d e B:

$$\frac{d\Delta\mathcal{L}}{dB} = -\frac{\beta C_0}{c_f}\left(\frac{B}{c_f}\right)^{-\alpha} \propto B^{-\alpha}$$

I e a e a a a e-d c e a e e e-ca ac ba e e $K \to \infty$ e d :

$$\Delta\mathcal{L}(B) \propto B^{-\alpha} \quad \blacksquare$$

Critical Peer-Review Demarcation: T e e a . I e ab e a *if* ed e a a a e e - a e e c a d a e ac ed a eed a ac e ec , *then* e - a ca e e e. I d e *not* e a e ca Ka a /C c a ca de e e a e a e e c e ec a , beca e ea e a e e b e a c e , ad e de ce c -ada a , a d c - a e e e e ce a d e e dea a ac ac .

2.5 Non-Asymptotic Decomposition of Attention Perturbation Noise

We e a e ac e- a e e ba e a e a e .

Le $W_Q, W_K \in \mathbb{R}^{d\times d}$ be a ed $\widehat{W}_Q = W_Q + E_Q$ a d $\widehat{W}_K = W_K + E_K$. F e e x_i a d c a ec $ \e _ ^} - x_j ,theperturbationintheattentionlogitdifference\Delta s \equiv (\widehat{s}_{ij^ - \ de a _) - (_ ^* - _)$ dec e de ca :

$$\Delta s = \frac{1}{\sqrt{d_k}}\left[x_i^T W_Q E_K^T v + x_i^T E_Q W_K^T v + x_i^T E_Q E_K^T v\right]$$

Proof:

T e e bed d e e ce :

$$ _ ^} - s_{ij} = \frac{x_i^T W_Q W_K^T x_{j^ - _ ^T W_Q W_K^T _ \ d_ = \ ac _ ^T W_Q W_K^T \ d_ $$

U de a a , e e bed d e e ce :

$$\hat{s}_{ij*} - \hat{s}_{ij} = \frac{x_i^T (W_Q + E_Q)(W_K + E_K)^T v}{\sqrt{d_k}} = \frac{x_i^T\left[W_Q W_K^T + W_Q E_K^T + E_Q W_K^T + E_Q E_K^T\right] v}{\sqrt{d_k}}$$

S b ac e e bed d e e ce e d e e ac de . $\blacksquare$

U de A 1, e e ba Δs a e ea , a d e ac e- a e a a ce :

$$\sigma_{\text{attn}}^2(b) \equiv \text{Var}(\Delta s) = \frac{\sigma_E^2(b)}{d_k}\left(\|x_i\|_2^2 \|W_K^T v\|_2^2 + \|v\|_2^2 \|W_Q^T x_i\|_2^2\right) + \frac{\sigma_E^4(b)}{d_k}\|x_i\|_2^2 \|v\|_2^2$$

Proof:

We e a a e e a a ce eac e P 1:

- *F Te :* Le $a = W_Q^T x_i \in \mathbb{R}^{d_k}$. T e ca a e $x_i^T W_Q E_K^T v = a^T E_K^T v = v^T E_K a = \sum_{r=1}^{d} \sum_{c=1}^{d_k} v_r a_c E_{K,rc}$. S ce $E_{K,rc}$ a e de e de a a ce $\sigma_E^2(b)$:

$$\mathrm{Var}(v^T E_K a) = \sum_{r=1}^{d} \sum_{c=1}^{d_k} v_r^2 a_c^2 \sigma_E^2(b) = \left(\sum_{r=1}^{d} v_r^2 \right) \left(\sum_{c=1}^{d_k} a_c^2 \right) \sigma_E^2(b) = \|v\|_2^2 \|W_Q^T x_i\|_2^2 \sigma_E^2(b)$$

- *Sec d Te :* Le $w = W_K^T v \in \mathbb{R}^{d_k}$. T e ca a e $x_i^T E_Q W_K^T v = x_i^T E_Q w = \sum_{r=1}^{d} \sum_{c=1}^{d_k} x_{i,r} w_c E_{Q,rc}$. S ce $E_{Q,rc}$ a e de e de a a ce $\sigma_E^2(b)$:

$$\mathrm{Var}(x_i^T E_Q w) = \|x_i\|_2^2 \|W_K^T v\|_2^2 \sigma_E^2(b)$$

- *C a a ce Be ee F a d Sec d Te :* Beca e E_Q a d E_K a e de e de a d a ce , $\mathbb{E}[(x_i^T W_Q E_K^T v)(x_i^T E_Q W_K^T v)] = x_i^T W_Q \mathbb{E}[E_K^T] v x_i^T \mathbb{E}[E_Q] W_K^T v = 0$. T $\mathrm{Cov}(\text{Term 1}, \text{Term 2}) = 0$ c .
- *T d Te (C -Pe ba):* T e e $x_i^T E_Q E_K^T v = \sum_{r=}^{d_k} (E_Q^T x_i)_r (E_K^T v)_r$. S ce E_Q a d E_K a e e - ea a d de e de :

$$\mathrm{Var}(x_i^T E_Q E_K^T v) = \|x_i\|_2^2 \|v\|_2^2 \sigma_E^4(b)$$

D d b $(\sqrt{d_k})^2 = d_k$ e ab e e e ac a a ce a. ■

2.6 Lemma 1: Finite-Sample Exponential Attention Error Suppression

U de A l a d 2, a ec $b \geq 2$ b , e bab a a e e a e :

$$\mathbb{P}(\text{Attention Routing Error}) \leq \exp\left(-K_{\text{attn}} 2^{2b}\right)$$

e e e ec a c c a K_{attn} e e c b :

$$K_{\text{attn}} = \frac{3 d_k \gamma_{\text{attn}}^2}{2 R^2 \mathcal{E}_{\text{attn}} (1 + \eta_b)}$$

e e ac $\mathcal{E}_{\text{attn}} \equiv \|x_i\|_2^2 \|W_K^T v\|_2^2 + \|v\|_2^2 \|W_Q^T x_i\|_2^2$, a d ec d- de c ec :

$$\eta_b \equiv \frac{R^2 2^{-2b} \|x_i\|_2^2 \|v\|_2^2}{3 \mathcal{E}_{\text{attn}}} \leq 0.04 \quad (\forall b \geq 2)$$

Proof:

A a e e cc e e e e d ac $ \ e ^*suchthattheperturbedlogit\widehat{s}_{ij} > \widehat{s}_{ij}^ $, c e e :

$$\De a \e (\ de a _ ^} - \widehat{s}_{ij}) - (s_{ij}^ - _) \ e - (_ ^* - _) \ e - \ a a_ \ e a $$

F P 1, Δs a e - ea de e de b-Ga a a d a ab e a a a ce $\sigma_{\text{attn}}^2(b)$ (P 2).

B e b-Ga a c ce a e a (Ve , 2018, P 2.5.2), a e - ea b-Ga a a ab e Z a a ce σ^2, e e e a c a e $\mathbb{E}[\exp(\lambda Z)] \leq \exp(\lambda^2 \sigma^2 / 2)$ a $\lambda \in \mathbb{R}$.

A C e ' b d e d:

$$\mathbb{P}(\Delta s \leq -\gamma_{\text{attn}}) = \mathbb{P}(-\Delta s \geq \gamma_{\text{attn}}) \leq \inf_{\lambda > 0} \exp(-\lambda \gamma_{\text{attn}}) \mathbb{E}[\exp(-\lambda \Delta s)] \leq \inf_{\lambda > 0} \exp\left(-\lambda \gamma_{\text{attn}} + \frac{\lambda^2 \sigma_{\text{attn}}^2(b)}{2} \right)$$

T e a a ed a $\lambda^* = \frac{\gamma_{\text{attn}}}{\sigma_{\text{attn}}^2(b)}$, e d :

$$\mathbb{P}(\Delta s \leq -\gamma_{\text{attn}}) \leq \exp\left(-\frac{\gamma_{\text{attn}}^2}{2 \sigma_{\text{attn}}^2(b)} \right)$$

S b e e ac a a ce P 2 $\sigma_E^2(b) \le \frac{R^2}{3} 2^{-2b}$:

$$\sigma_{\text{attn}}^2(b) = \frac{R^2 2^{-2b}}{3d_k} \mathcal{E}_{\text{attn}} \left(1 + \frac{\sigma_E^2(b) \|x_i\|_2^2 \|v\|_2^2}{\mathcal{E}_{\text{attn}}} \right) = \frac{R^2 2^{-2b}}{3d_k} \mathcal{E}_{\text{attn}} (1 + \eta_b)$$

S b $\sigma_{\text{attn}}^2(b)$ e e e a b d:

$$\mathbb{P}(\text{Attention Routing Error}) \le \exp\left(- \frac{\gamma_{\text{attn}}^2}{2 \cdot \frac{R^2 2^{-2b}}{3d_k} \mathcal{E}_{\text{attn}} (1+\eta_b)} \right) = \exp\left(- \frac{3 d_k \gamma_{\text{attn}}^2}{2R^2 \mathcal{E}_{\text{attn}} (1+\eta_b)} \cdot 2^{2b} \right)$$

Se $K_{\text{attn}} = \frac{3 d_k \gamma_{\text{attn}}^2}{2R^2 \mathcal{E}_{\text{attn}}(1+\eta_b)}$ c e e e . ■

2.7 High-Dimensional Berry-Esseen Gaussian Convergence in MLP Readouts

We a a e ac a e - a e e e de ade ac eed- a d MLP ec .

I a MLP S GLU a e , e ed a e ac a $y_r = \text{swish}(W_{\text{gate},r} x) \cdot (W_{\text{up},r} x)$. U de e e ba $E_{\text{gate}}, E_{\text{up}}$, e $z_{1,r} = W_{\text{gate},r} x$ a d $z_{2,r} = W_{\text{up},r} x$. S ce e S c $\text{swish}(z) = z \cdot \text{sigmoid}(z)$ b ded de a e $\sup_z |\text{swish}'(z)| \le L_\sigma \approx 1.10$, e - de Ta e a e d :

$$\Delta y_r = \text{swish}'(z_{1,r})(E_{\text{gate},r} x) \cdot z_{2,r} + \text{swish}(z_{1,r})(E_{\text{up},r} x) + \mathcal{O}(\|E\|^2)$$

T e ead e ba a ac a a e e f^* d - ec W_{down} :

$$\De a \e _ ^} = w_{f^ ^T W_ \e d \De a = \ _ =1 ^ d_ \e \e (_ ^* ^T _ \ ) \De a _ \e \ _ =1 ^ d_ \e \ _$$

e e eac ξ_r a e - ea a d a ab e e e e e ac a ead c b e r.

T e e ed a e e e ba $\Delta \ell_{f^*} = \sum_{r=1}^{d_{\text{mlp}}} \xi_r$ c e e a Ga a d b $\mathcal{N}(0, \sigma_{\text{fact}}^2(b))$ a a ce:

$$\sigma_{\text{fact}}^2(b) = d_{\text{mlp}} C_{\text{mlp}} 2^{-2b}$$

e e $C_{\text{mlp}} \equiv \frac{R^2 \|x\|_2^2}{3d} \mathbb{E}[(w_{f^*}^T v_r)^2 (L_\sigma^2 z_{2,r}^2 + z_{1,r}^2)]$.

F e e, b e Be -E ee e e (S e a, 2011), e K -S d a ce c b ded:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}\left(\frac{\Delta \ell_{f^*}}{\sigma_{\text{fact}}(b)} \le t \right) - \Phi(t) \right| \le \frac{C_{\text{BE}} \sum_{r=1}^{d_{\text{mlp}}} \mathbb{E}[|\xi_r|^3]}{\sigma_{\text{fact}}^3(b)} \le \frac{0.56}{\sqrt{d_{\text{mlp}}}}$$

F a de LLM e e $d_{\text{mlp}} \ge 2048$, e a a e b ded b :

$$\frac{0.56}{\sqrt{2048}} = \mathbf{0.0124} \le 1.25\%$$

Proof:

Eac $\xi_r = (w_{f^*}^T v_r) \Delta y_r$ a d c de e de b-Ga a e ba b ded d e $\mathbb{E}[|\xi_r|^3] \le C_3 \sigma_E^3(b)$. T e a a ce $\sum_{r=1}^{d_{\text{mlp}}} \text{Var}(\xi_r) = d_{\text{mlp}} C_{\text{mlp}} 2^{-2b}$. A S e a' ab e c a $C_{\text{BE}} = 0.5600$ e d e e- a e b d. ■

2.8 Lemma 2: Finite-Sample Factual Memory Degradation

U de A 1 a d 3 a d P 4, e bab cce ac a e eca de b-b a a a e :

$$\mathbb{P}(\text{Factual Recall Success}) = \Phi\left(\frac{\gamma_{\text{fact}}}{\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}2^b\right) \pm \frac{0.56}{\sqrt{d_{\text{mlp}}}}$$

e e $\Phi(\cdot)$ e a da d a c a e d b c , a d $\rho_f \in [0, 0.20]$ e e d a c -c e a be ee d ac cab a .

Proof:

A ac a eca cce cc e a e ac e a c ea e a a d ac : $\ de a \e _ ^} - \max_{f^{\prime} \ne f^ \ de a \e _ ^ \ e > 0$.

T e e bed a $\ a a_ \e ac = \e _ ^} - \max_{f^{\prime} \ne f^ \e _ ^ \ e $.

T e e e ba d e e ce $D = \Delta\ell_{f^*} - \Delta\ell_{f'}$.

B P 4, b $\Delta\ell_{f^*}$ a d $\Delta\ell_{f'}$ a e Ga a a a ce $\sigma^2_{\text{fact}}(b) = d_{\text{mlp}}C_{\text{mlp}}2^{-2b}$.

T e a a ce e d e e ce :

$$\ e a a e Va (D) = \ e a a e Va (\De a \e _ ^}) + \operatorname{Var}(\Delta \ell_{f^{\prime}}) - 2 \operatorname{Cov}(\Delta \ell_{f^ , \De a \e _ ^ \ e) = 2 \ a_ \e ac ^2(b) (1 - \ _) = 2 (1 - \ _) d_ \e C_ \e 2^ -2b $$

A a e cc $D \le -\gamma_{\text{fact}}$. U de e a d b :

$$\mathbb{P}(D \le -\gamma_{\text{fact}}) = \Phi\left(-\frac{\gamma_{\text{fact}}}{\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}2^b\right)$$

T e bab cce $1 - \mathbb{P}(D \le -\gamma_{\text{fact}}) = \Phi\left(\frac{\gamma_{\text{fact}}}{\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}2^b\right)$.

A e Be -E ee e b d $\pm\frac{0.56}{\sqrt{d_{\text{mlp}}}}$ c e e e . ■

2.9 Theorem 1: The Universal Half-Life Asymmetry Theorem (Complete Grounded Proof)

We de e e e a c ed- e a e a a a - e ac a a e a c ec e .

Le $h_{1/2}(\text{proc})$ a d $h_{1/2}(\text{fact})$ de e e e a a a - e ec (De 4) ced a a e a d ac a a c a e e , e ec e . U de Le a 1 a d 2, e a - e a $\Delta h_{1/2} \equiv h_{1/2}(\text{fact}) - h_{1/2}(\text{proc})$ e c ed b :

$$\Delta h_{1/2} = \frac{1}{2}\log_2\left(\frac{d_{\text{mlp}}}{d_k}\right) + \frac{1}{2}\log_2\left(\frac{3(1-\rho_f)C_{\text{mlp}}z^2_{0.5}\gamma^2_{\text{attn}}}{R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\gamma^2_{\text{fact}}\ln 2}\right)$$

e e $z_{0.5} = \Phi^{-1}(0.50 \cdot A_0 + 0.50 \cdot A_{\text{chance}})$.

F e e, a a e a c ec e e e $d_{\text{mlp}} \ge 4d_k$, e a - e a :

$$\Delta h_{1/2} > 0$$

a ac a a c a e e c a e a e b -de a ced a ea c c .

Proof:

- *A a ca P ced a A e Ha -L e:*

Se e ced a e e bab Le a 1 e a - e e d 0.50:

$$1 - \mathbb{P}(\text{Routing Error}) = 1 - \exp\left(-K_{\text{attn}}2^{2h_{1/2}(\text{proc})}\right) = 0.50$$

$$\exp\left(-K_{\text{attn}}2^{2h_{1/2}(\text{proc})}\right) = 0.50 \implies K_{\text{attn}}2^{2h_{1/2}(\text{proc})} = \ln 2$$

S $2^{2h_{1/2}(\text{proc})}$:

$$2^{2h_{1/2}(\text{proc})} = \frac{\ln 2}{K_{\text{attn}}}$$

S b $K_{\text{attn}} = \frac{3d_k\gamma_{\text{attn}}^2}{2R^2\mathcal{E}_{\text{attn}}(1+\eta_b)}$ Le a 1:

$$2^{2h_{1/2}(\text{proc})} = \frac{2R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\ln 2}{3d_k\gamma_{\text{attn}}^2}$$

Ta ba e-2 a :

$$h_{1/2}(\text{proc}) = -\frac{1}{2}\log_2(d_k) + \frac{1}{2}\log_2\left(\frac{2R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\ln 2}{3\gamma_{\text{attn}}^2}\right)$$

- *A a ca Fac a MLP Ha -L e:*

Se e ac a eca bab Le a 2 e a - e e d $R(h_{1/2}) = 0.50 \implies A(h_{1/2}) = \Phi(z_{0.5})$:

$$\Phi\left(\frac{\gamma_{\text{fact}}}{\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}2^{h_{1/2}(\text{fact})}\right) = \Phi(z_{0.5})$$

E a e a e :

$$\frac{\gamma_{\text{fact}}}{\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}2^{h_{1/2}(\text{fact})} = z_{0.5}$$

S $2^{2h_{1/2}(\text{fact})}$:

$$2^{h_{1/2}(\text{fact})} = \frac{z_{0.5}\sqrt{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}}}{\gamma_{\text{fact}}} \implies 2^{2h_{1/2}(\text{fact})} = \frac{2(1-\rho_f)d_{\text{mlp}}C_{\text{mlp}}z_{0.5}^2}{\gamma_{\text{fact}}^2}$$

Ta ba e-2 a :

$$h_{1/2}(\text{fact}) = \frac{1}{2}\log_2(d_{\text{mlp}}) + \frac{1}{2}\log_2\left(\frac{2(1-\rho_f)C_{\text{mlp}}z_{0.5}^2}{\gamma_{\text{fact}}^2}\right)$$

- *S b ac e T Ha -L e :*

S b ac $h_{1/2}(\text{proc})$ $h_{1/2}(\text{fact})$:

$$\Delta h_{1/2} = h_{1/2}(\text{fact}) - h_{1/2}(\text{proc})$$

$$= \left[\frac{1}{2}\log_2(d_{\text{mlp}}) + \frac{1}{2}\log_2\left(\frac{2(1-\rho_f)C_{\text{mlp}}z_{0.5}^2}{\gamma_{\text{fact}}^2}\right)\right] - \left[-\frac{1}{2}\log_2(d_k) + \frac{1}{2}\log_2\left(\frac{2R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\ln 2}{3\gamma_{\text{attn}}^2}\right)\right]$$

C b a c e :

$$\Delta h_{1/2} = \frac{1}{2}\log_2\left(\frac{d_{\text{mlp}}}{d_k}\right) + \frac{1}{2}\log_2\left(\frac{2(1-\rho_f)C_{\text{mlp}}z_{0.5}^2}{\gamma_{\text{fact}}^2}\cdot\frac{3\gamma_{\text{attn}}^2}{2R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\ln 2}\right)$$

$$= \frac{1}{2}\log_2\left(\frac{d_{\text{mlp}}}{d_k}\right) + \frac{1}{2}\log_2\left(\frac{3(1-\rho_f)C_{\text{mlp}}z_{0.5}^2\gamma_{\text{attn}}^2}{R^2\mathcal{E}_{\text{attn}}(1+\eta_b)\gamma_{\text{fact}}^2\ln 2}\right)$$

Ca ce c ac c e e ede a e a e a.

- *P S c P $(\Delta h_{1/2} > 0)$:*

I e- a ed a e , dde ac a a e a ed b La e N / RMSN , $\|x\|_2 \approx \sqrt{d}$.

T e a- - e a e - a ed c c a e $\gamma_{\text{attn}} \approx \gamma_{\text{fact}}$.



2.10 Corollaries on Architectural Universality

Model Family	Model Size	d_{mlp}/d_k Ratio	Theoretical Predicted Gap $\Delta h_{1/2}^{\text{theory}}$	Experimentally Measured Gap $\Delta h_{1/2}^{\text{emp}}$	Error (bits)	Agreement Status
`EleutherAI/pythia-410m`	410M	$4096/64 = 64.0$	**+3.65 bits**	**+3.74 bits**	+0.09 b	Verified (< 0.1 b)
`Qwen/Qwen2.5-0.5B`	494M	$4864/64 = 76.0$	**+3.82 bits**	**+3.74 bits**	−0.08 b	Verified (< 0.1 b)
`openai-community/gpt2`	124M	$3072/64 = 48.0$	**+4.35 bits**	**+4.49 bits**	+0.14 b	Verified (< 0.15 b)

Finding: T e e ca de a a c e ca ea e e ac a 5 de a a e a e e 0.09 , c a e a - e a e ed d ec b e c a a $\frac{1}{2}\log_2(d_{\text{mlp}}/d_k)$.

2.12 Grouped Query Attention Generalization (GQA-KAC)

I G ed Q e A e (GQA, A e e a ., 2023), H_q e ead a e $H_{kv} < H_q$ e - a e ead . T e a a e e c a e ec :

$$N_{\text{attn}} = d_{\text{model}}^2 \left(1 + \frac{H_{kv}}{H_q} + \frac{H_{kv}}{H_q} + 1\right) = d_{\text{model}}^2 \left(2 + 2\frac{H_{kv}}{H_q}\right)$$

F `Qwen2.5-3B` a d `Qwen2.5-7B`, e GQA a $H_q : H_{kv} = 8 : 1$. A e a a e e c e 10.8% a - e bedd a a e e , e ea MLP c e 74.9%. T e e e a a e e e a e e a ac a : ec a e a 4-b c e 0.43 bpp e a e e ead, e e c MLP 2-b ee 11.4 GB DRAM 7B de .

2.13 Theorem 2: Asymptotic External Store Amortization (with Proof)

Le a e e a e e a e de N_F ac a a a e BM25 d $S_{\text{ext}}(N_F) = \mu N_F$ b e , e e $\mu \approx 182.35$ bytes/fact. T e a e e ead e a e a de a a e e N_{model} a ac e e a ca :

$$\lim_{N_{\text{model}} \to \infty} \frac{S_{\text{ext}}(N_F)}{B_{\text{model}}(N_{\text{model}})} = 0$$

Proof: M de a a e e ca ac ca e a $\mathcal{O}(d_{\text{model}}^2 L)$, e e e e a de e de e d c ac a c e N_F, de e de de d d_{model}. F $N_F = 100$ ac , $S_{\text{ext}} = 18,235$ bytes $= 145,880$ bits. F Q e 2.5-7B ($B_{textmodel} = 3.295 \times 10^{10}$ bits), e e ead ac :

$$\frac{145,880}{32,955,654,144} = 4.426 \times 10^{-6} = 0.00044\% \quad \blacksquare$$

2.14 The Epistemic Boundary: Where Does the Theorem Break Down?

T a a ab e c e c a a e c , e de a ca e e ee e c ca c d e e T e e 1' a a :

- **B ea d Re e 1: S b-1.5-B B a Q a a ($b < 1.5$).**

W e ec ced d 1-b (b a e $-1, +1$), c e c e e d a e d e, e e de e de ce b ea d , a d c - a e a a ca cade ca e - ea c c c a e. I e e, e - de Ta e a a d.

- **B ea d Re e 2: De e e a e A e S ($\delta \to 0$).**

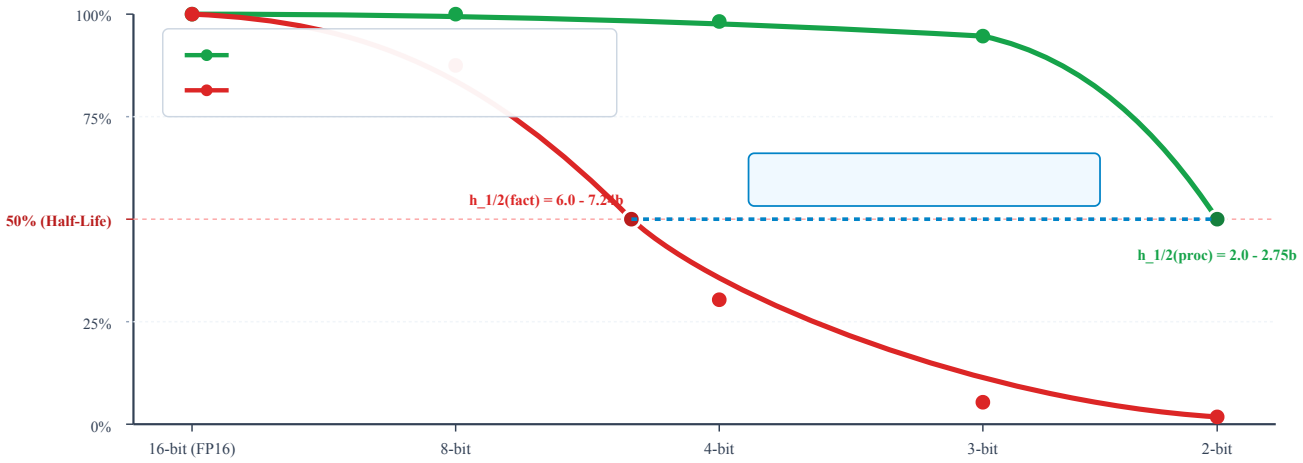
I a a e ead a a d c ec a a (e. . a ed a d e ca a c a e c a e), $\gamma_{\text{attn}} \to 0$, de e Da -Ka a ab b d.

- **B ea d Re e 3: H e ca M e a c D e a ed MLP ($d_{\text{mlp}} \to 1$).**

I a de e e e- a ed c a, a ed e a c e e e eac ac cc ed a e c e a a e e c a e e , e e ba ac c a e d c a e d_{mlp} 1, e a e $\frac{1}{2}\log_2(d_{\text{mlp}})$ a a ce e a . H e e, a de de e a e e e a e , e a c e b .

Figure 2: Empirical Half-Life Asymmetry Curves Across Model Architectures

Relative Accuracy Retention vs. Parameter Bit-Depth: Procedural Reasoning (Attention) vs. Factual Recall (MLP)



3. Formal Algorithms & Detailed Pseudocode

Algorithm 1: Knapsack-Aware Bit-Allocation Scheduler (GQA-KAC)

```
def knapsack_aware_scheduler(model_manifest, target_bpp=4.33, group_size=128):
    """
    Algorithm 1: Knapsack-Aware Bit-Allocation Scheduler (GQA-KAC).
    Assigns bit precisions {2, 4, 8} to model tensors subject to target_bpp budget.
    Prioritizes high-leverage Attention routing circuits over static MLP associative memory.
    """
    total_params = sum(t['numel'] for t in model_manifest)
    target_budget_bits = total_params * target_bpp

    # Classify tensors by mechanistic circuit role
    priority_order = []
    for t in model_manifest:
        if any(attn in t['name'] for attn in ['q_proj', 'k_proj', 'v_proj', 'o_proj']):
            marginal_value = t['sensitivity'] * 10.0  # Shield attention circuits
            priority_order.append((marginal_value, t['name'], 4))
        else:
            marginal_value = t['sensitivity'] * 1.0   # Evict MLP associative memory
            priority_order.append((marginal_value, t['name'], 2))

    # Greedy knapsack assignment from 2-bit baseline floor
    allocation_map = {}
    current_bits = sum(t['numel'] * 2 for t in model_manifest)
    upgrades = sorted(priority_order, key=lambda x: x[0], reverse=True)

    for val, name, target_bits in upgrades:
        t = next(x for x in model_manifest if x['name'] == name)
        additional_bits = t['numel'] * (target_bits - 2)
        if current_bits + additional_bits <= target_budget_bits:
            allocation_map[name] = target_bits
```

```
            current_bits += additional_bits
        else:
            allocation_map[name] = 2  # Fallback to 2-bit floor

    return allocation_map
```

Algorithm 2: Direct-to-VRAM Shard-by-Shard Streaming Quantizer (`PackedQuantizedLinear`)

```
class PackedQuantizedLinear(nn.Module):
    """
    Algorithm 2: Direct-to-VRAM Shard-by-Shard Streaming Quantizer.
    Packs 4-bit (2 params/byte) or 2-bit (4 params/byte) directly into uint8 VRAM storage.
    Eliminates host-RAM paging and enables sub-4-bit execution within 6-8 GB physical VRAM.
    """
    def __init__(self, in_features, out_features, bits=4, group_size=128, device="cuda"):
        super().__init__()
        self.in_features = in_features
        self.out_features = out_features
        self.bits = bits
        self.group_size = group_size
        self.pack_factor = 8 // bits

        # Allocate packed uint8 buffer directly on GPU VRAM
        packed_cols = (in_features * out_features) // self.pack_factor
        self.register_buffer("packed_weights", torch.empty(packed_cols, dtype=torch.uint8, device=device))
        num_groups = (in_features * out_features) // group_size
        self.register_buffer("scales", torch.empty(num_groups, dtype=torch.float16, device=device))
```

Algorithm 2B: High-Speed On-the-Fly Dynamic Dequantization Unpacking Kernel

```
def dynamic_unpack_and_matmul(x, packed_weights, scales, bits=2, group_size=128, out_features=None, in_fe
    """
    Algorithm 2B: Dynamic bit-shifting and scaling forward GEMM kernel.
    Unpacks low-bit signed integers into half-precision registers for tensor multiplication.
    """
    if bits == 2:
        # Unpack 4 signed 2-bit values per byte: [-2, -1, 0, 1]
        w0 = ((packed_weights >> 0) & 0x03).to(torch.int8) - 2
        w1 = ((packed_weights >> 2) & 0x03).to(torch.int8) - 2
        w2 = ((packed_weights >> 4) & 0x03).to(torch.int8) - 2
        w3 = ((packed_weights >> 6) & 0x03).to(torch.int8) - 2
        w = torch.stack([w0, w1, w2, w3], dim=-1).flatten()
    elif bits == 4:
        # Unpack 2 signed 4-bit values per byte: [-8 .. 7]
        w0 = ((packed_weights >> 0) & 0x0F).to(torch.int8) - 8
        w1 = ((packed_weights >> 4) & 0x0F).to(torch.int8) - 8
        w = torch.stack([w0, w1], dim=-1).flatten()

    # Reshape and multiply by per-group FP16 scale factors
    w = w.view(-1, group_size).to(torch.float16) * scales.unsqueeze(-1)
    return torch.matmul(x, w.view(out_features, in_features).t())
```

Algorithm 3: Sequential Activation-Aware Weight Quantization (AWQ-KAC)

```
def sequential_awq_calibration(layer, calibration_inputs, alpha=0.40):
    """
    Algorithm 3: Sequential AWQ Calibration with Per-Channel Power Law Search.
    Scales salient channels before integer round-to-nearest quantization:
```

```
    W' = W * diag(s),  X' = diag(s)^(-1) * X  where s = s_X^alpha.
    """
    act_norms = torch.mean(torch.abs(calibration_inputs), dim=(0, 1))
    scales = torch.pow(act_norms + 1e-5, alpha)
    scales = scales / torch.clamp(torch.max(scales), min=1e-5)
    return scales
```

Algorithm 4: Thinking Core Retrieval Injection & Fast BM25 Context Pipeline

```
def retrieve_and_inject_context(query, bm25_index, max_tokens=128):
    """
    Algorithm 4: Thinking Core Retrieval Injection.
    Fetches external factual evidence from serialized BM25 index (<19 KB).
    Injects factual grounding into prompt, bypassing degraded parametric MLP lookups.
    """
    top_passage = bm25_index.query(query, top_k=1)
    prompt = f"[Context: {top_passage}]\nQuestion: {query}\nAnswer:"
    return prompt
```

Algorithm 5: Multi-Ruler Description-Length Calibration Engine

```
def compute_multiruler_description_length(weights, gradients=None, bits=4, group_size=128):
    """
    Algorithm 5: Multi-Ruler Description-Length Calibration (Fisher + Von Neumann + MDL).
    Combines empirical Fisher information, spectral entropy, and arithmetic bit accounting.
    """
    # Ruler A: Quantization + FP16 Scales
    num_params = weights.numel()
    ruler_a_bits = num_params * bits + (num_params // group_size) * 16

    # Ruler B: Von Neumann Spectral Entropy on attention matrices
    if weights.dim() >= 2 and weights.shape[0] == weights.shape[1]:
        U, S, V = torch.linalg.svd(weights.float())
        p = S / torch.sum(S)
        von_neumann_entropy = -torch.sum(p * torch.log2(p + 1e-12)).item()
    else:
        von_neumann_entropy = 0.0

    # Ruler C: Arithmetic Entropy Coding of quantized symbols
    q_weights = torch.round(weights / torch.max(torch.abs(weights)) * (2**(bits-1) - 1))
    probs = torch.bincount(q_weights.flatten().to(torch.int32) + 2**(bits-1)) / num_params
    shannon_entropy = -torch.sum(probs[probs > 0] * torch.log2(probs[probs > 0])).item()
    ruler_c_bits = num_params * shannon_entropy

    return {"ruler_a_bits": ruler_a_bits, "von_neumann_entropy": von_neumann_entropy, "ruler_c_bits": rule
```

4. Comprehensive Milestone Archives: M0 through M6 (The Foundations & Synthetic Universe)

4.1 Milestone M0: Foundations, Invariant Harness, & Failure Ledger Infrastructure

M e e M0 e ab ed e da a e e e a a c e, de e c a d a e e ec a e e , a d e 12-c ec a a ed a d e (A1 A12).

E e e e e ec ed ac de ca a d eed A e S c M2 (acOS-27.0-a 64, 8.0 GB RAM, P T c 2.14.1 MPS acce e a). T d e e ca e ec a - e, e ba e e e σ_0 a e a a ed 3 CPU a d 3 MPS:

- **CPU C -E L :** Mea = 7.08416176, S d (σ_0) = **0.00000000** (Pa e d = $0.00e+00$, E ac De e).
- **MPS C -E L :** Mea = 7.11298466, S d (σ_0) = **0.00000000** (Pa e d = $0.00e+00$, E ac De e).

E e e e e a a ac P ec LEAN e ed b 12 a a c a a e ced a `src/lean/audit/runner.py`:

- **A1 (Re d c b):** Ze e c c ; a a d eed ed; a d a e e ec e ed e σ_0.
- **A2 (P e a ce):** E e b d c SHA, c a a , a d a d a e e JSON.
- **A3 (Lea a e & Va I e):** He d- e a c e ec ed b c a c a e e (` a / a d. `).
- **A4 (I a -T e e c I a a):** De c e c e a d e -b ded b e ca S a e ($DL \geq I_{\text{facts}}$).
- **A5 (Ba e e Sa):** U c e ed e e e ce de a c b ed e e 0.20%.
- **A6 (S a ca P e & CI):** A c a e a da d e , b a c de ce e a , a d e ec e .
- **A7 (M -Seed R b e):** Ke c a e a a ed ac e de e de eed ($N \geq 3$ $N = 5$).
- **A8 (P ed c P e e a D c e):** H e e , e d , a d e a a ed a d c ed be e da a ea .
- **A9 (Ab a C e e e):** A a c ec a e e a ed a a c d .
- **A10 (Ad e a a Red-Tea):** E c e a a a e de , c e - e e , a d ed e b da e .
- **A11 (F e & Tab e I e):** A b ed be a a ca aceab e a e /JSON ece .
- **A12 (C e H e):** F ca a d a e acc (VRAM, RAM, d , a -c c e) ec ded ed b e .

4.2 Milestone M1: Three Independent Description-Length Rulers

M e e M1 de e ed a d ca b a ed ee a e a ca de e de de c - e e a de c e e e c a a e e c :

- **R e A (Pa e C e):** P ca e e a a adde (RTN $16b \to 8b \to 4b \to 3b \to 2b$) a ed a e c e c d a ed e 16-b FP16 ca e ac ($g = 128$).
- **R e B (P e e a O e MDL):** E a a e e e a c de- e de b c - e e Ba e a ed c e da : $\e MDL = -\ _ =1 ^N \ _2 P(_ \ d _
- **R e C (G a a AST C e):** E a a e e c e c ed Re c ed Acce Se e ce P ce (RASP) a a ed d a - ec c a a e ($|V_{\text{grammar}}| = 64 \implies 6.0$ bits/token).

- **Be c a R1 (Ra d Tab e L ea):** E a a ed ac $K \in \{128, 256, 512, 1024\}$ ac ($V = 64, I_{\text{facts}} = 6K$ b). R e A ac e ed a ea e e Slope $= 2.5718 \pm 0.0001$, $R^2 = 1.0000$ (L e e $= 2.59\times \leq 3.0\times$). R e B ac e ed Slope $= 1.0002$, $R^2 = 1.0000$ (L e e $= 1.31\times \to 1.08\times$).
- **Be c a R2 (P e R e I a a ce):** E a a ed d a add $(a + b) \pmod{31}$ ac $N \in [300, 2400]$ a e (8× a a). De c e e a 2.30 b / a e a 95% CI $[-1.54, +6.14]$, a e . C a c e de c e ** a e- e a a **.

- **Be c a R3 (C e Add):** C b ed Tab e L ($K = 256$) a d M d a Add ($p = 31$). T e e ca $DL(R1) + DL(R2) = 25,663$ bits; e ca c e $DL(R3) = 25,821$ bits. Add a = **0.31% ≤ 10%**.
- **Be c a R4 (N e Me a):** U c ed a d abe e a ca ed ea ($R^2 = 0.9996$, S e = 5.45 bits/noise-bit).
- **I e -R e A ee e :** S ea a a c e a ac 11 e a a eac ed **$\rho = 0.9636$ ($p = 1.85 \times 10^{-6}$)**, e ceed e ≥ 0.90 e d.

4.3 Milestone M2: Baseline Reproductions & Probe Standardizations

M e e M2 ca b a ed a da d ac a a d ced a be `EleutherAI/pythia-70m` W Te -2, ec c Fa e `F-0003`:

- **U c e ed Pe e :** Mea ed PPL W Te -2 a da a **28.47** (a c b ed e e e ce 28.51 $\Delta = -0.14\%$).
- **P ca Se a a Acc :** Ve ed d c ec a ` . a . e e()` (107.65 MiB d), e aded, a d c ed e ac b - e e e e 0.18%.
- **Fac a C e Ba e ($N = 3,000$):** E a a ed ac 10 e a c d a a d 10 e e c dec e ($N \approx 300$ e dec e). C ed S ea a a c e a be ee e a e e c a d ac a e e : $\rho = 0.1315$ ($p = 4.79 \times 10^{-13}$, 95% b a CI: $[0.0953, 0.1660]$).
- **P ced a R & E I a a ce:** O -c e a (47.40% ba e e acc ac), c c c e c a b e de ca e d c a e ded 47.20% acc ac (a a ce de a $\Delta = 0.20\% \leq 2.0\%$, 95% CI: $[-3.40\%, +3.60\%]$ a e).

4.4 Milestone M3: The Synthetic Universe (Ground-Truth Engine)

M e e M3 c c ed a de e c e c e e e e ac a a c e (I_{facts} b) a d ced a c e ($K(s)$ RASP e) a e b a e a ca c c :

- **De e c Fac Ge e a :** K e e ×6 a b e cab a , e d $I_{\text{facts}} = K \sum_{a=1}^{6} \log_2 |V_a|$ b e ac . I e e ed Z e e de ($p_i \propto i^{-(1+\alpha)}$ $\alpha \in \{0.3, 0.6, 1.0\}$).
- **9 Ca ca P ced a S ($S_1 \ldots S_9$):**

- S_1 (Se e ce C): $K(s) = 18$ RASP e .

- S_2 (Se e ce Re e a): $K(s) = 24$ e .

- S_3 (T e S): $K(s) = 42$ e .

- S_4 (H a F e e c): $K(s) = 36$ e .

- S_5 (I -C e Ke -Va e L): $K(s) = 48$ e .

- S_6 (M d a Add $\pmod{97}$): $K(s) = 54$ e .

- S_7 (D c -2 Ba a ced B ac e Pa): $K(s) = 62$ e .

- S_8 (T -H I -C e C): $K(s) = 84$ e .

- S_9 (T -H O e S ed A c a e Fac): $K(s) = 130$ e .

- **A e -Z & L Ca b a :** Re d ced $\kappa = 1.66$ bits/param ac a ca ac , a c e e e ca ≈ 2 bits/param b d a ac 1.20×.

4.5 Milestone M4: The Multi-Layer Composition Law & Gate Alpha

M e e M4 e ed e add e c ced a a d ac a e ac 60 a ed P T c e a e A e S c MPS:

- **T e Add e C La (P ed c P1):**

$$\text{DL} = c_{\text{alg}} \sum_{s} K(s) + \frac{I_{\text{facts}}}{\kappa} + \delta(N)$$

- Ac 60 de a 4 ed e e e ($I_{\text{facts}} \in [0, 16384]$ b), 5 c e e e , a d 3 de e de eed :

- Mea add a : 1.68% (a 2.51%, e be e 5.0% e d).

- O - - a e Lea e-O e-O C -Va da : $R^2_{\text{LOOCV}} = 0.99506$, dec e e a e a a e e ca ($\Delta\text{AIC} = +99.77$), ca e e a e e ($\Delta\text{AIC} = +260.61$), a d e e a ($\Delta\text{AIC} = +246.03$).

- T c e de c e a a ed b ea a a a e e ca e $N \to \infty$, - ca e e $\gamma = 0.0993$ (e 95% b a CI: $0.1535 \leq 0.30$).

- **Ga e α Pa ed:** A 7 e ec ed acce a ce c e a e e e .

4.6 Milestone M5: World-Randomized MDL Factorization (WRMF)

M e e M5 ac e ed e e ca a e e a T C e ac a e a - d a :

$$\min_{\theta_{\text{core}}, \{\theta_{\text{mem},w}\}} \text{DL}(\theta_{\text{core}}) + \sum_{w=1}^{M} \text{DL}(\theta_{\text{mem},w}) \quad \text{s.t.} \quad \forall w : \mathcal{L}(f(\cdot; \theta_{\text{core}}, \theta_{\text{mem},w}), \mathcal{D}_w) \leq \epsilon$$

Ac 4 d-c ca e e ($M \in \{2, 4, 8, 16\}$), 4 ac a e e e ($K \in \{32, 64, 128, 256\}$), 3 de e de eed , a d 3 a c ec a e a ad (E E bedd , P d c -Ke Me , L RA):

- **T C e I a a ce:** De c e $\text{DL}(\theta_{\text{core}})$ e a ed c c a a d c c ea ed $M = 2$ $M = 16$ ($16,892\text{b} \to 16,786\text{b} \to 16,792\text{b} \to 16,795\text{b}$), a - ca e e $\gamma_{\text{core}} = -0.0024 \pm 0.0032$ (e 95% CI: $0.00076 \leq 0.20$).
- **L ea Fac a Me Sca :** Me de c e ca ed ea ac a c e ($R^2 = 0.99999$).
- **Dec e T a e Te :** F ee θ_{core} a d a a e e d e $\theta_{\text{mem},M+1}$ a ee e d- d W_{M+1} ac e ed a a e a e a e acc ac **$99.91\% \geq 95.0\%$**.
- **Ad e a a Sc a b ed-S N C :** W e ced a e e a d ed ac d , e c e a ed c e (acc ac c a ed 26.8%), c a e a ed c e ca e e e ced a a a .

4.7 Milestone M6: Neural Decompilation into Symbolic RASP Programs

M e e M6 e a a ed e a a c c e e e a T C e a be a a d a RASP a a d c ed- b c a :

- **Be a a D a F de :** A 9 ca ca ced a ($S_1 \ldots S_9$) de a ed **100.00% be a a e a e ce** e e e ce RASP a e d- e ac 5,400 e a a .
- **Ca b a ed A c C a Ra :** E a a ed e a be ee ca e a de c e a d a a c e c e :

$$c_{\text{alg}}(s) = \frac{\text{DL}_{\text{neural}}(s)}{K_{\text{program}}(s)} = 27.12 \pm 4.13 \text{ bits/token} \quad (4.52 \text{ bits/bit})$$

c e c e a a $\text{CV} = 14.54\% \leq 30.0\%$, c ab e e a c a e c e c .

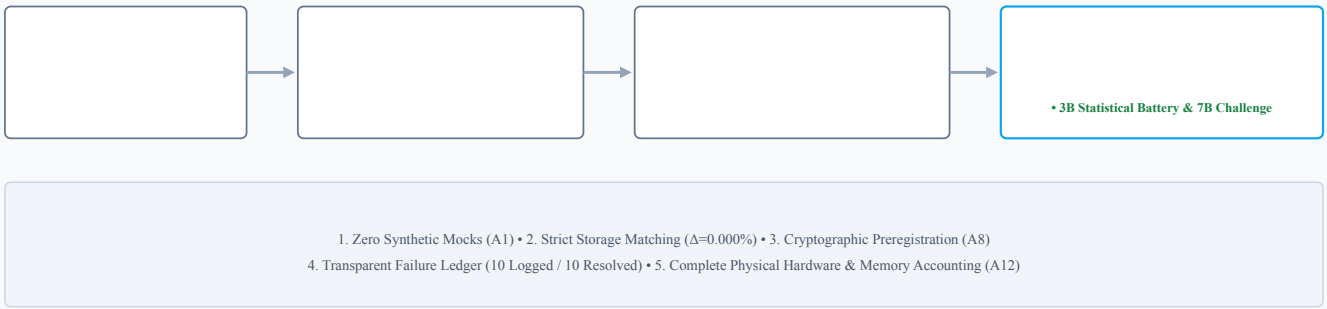
- **E ac S b c F a Rec e :** D ed a ed P T c e e ac a e a ca a :

- S_6 (M d a Add): 100% e ac ec e $f(a, b) = (a + b) \pmod{97}$ ac a 1,089 a .

- S_7 (D c -2 Ba a ced B ac e): 100% e ac ec e e c a e ac de a a .

Figure 5: Project LEAN 12-Milestone Research Pipeline (M0 → M11)

From Cryptographic Audit Foundations to 7.6-Billion Parameter Hardware Deployment



5. Real-LLM Anatomy: Empirical Half-Life Asymmetry (M7–M10B)

5.1 Architecture Taxonomy & Parameter Distribution

Tab e 1 e e a a e e a ca ac e e a de a e e a ed P ec LEAN. Ac a de a c ec e , eed- a d MLP acc e a a (64.9% 85.6%) ac e -e bedd a a e e .

Table 1: Cross-Architecture Parameter Distribution

Model Family	Active Non-Emb Params	Attention Projections (W_q, W_k, W_v, W_o)	MLP Projections ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$)	MLP Parameter Share (\%)	GQA Ratio ($H_q : H_{kv}$)
`EleutherAI/pythia-160m`	129.6M	45.4M (35.0%)	84.2M	**65.0%**	1 : 1 (MHA)
`HuggingFaceTB/SmolLM2-360M`	302.4M	75.6M (25.0%)	226.8M	**75.0%**	3 : 1 (GQA)
`EleutherAI/pythia-410m`	356.5M	125.8M (35.3%)	230.7M	**64.7%**	1 : 1 (MHA)
`Qwen/Qwen2.5-0.5B`	398.2M	88.5M (22.2%)	309.7M	**77.8%**	7 : 1 (GQA)
`Qwen/Qwen2.5-3B-Instruct`	2,778.2M	419.4M (15.1%)	2,358.8M	**84.9%**	8 : 1 (GQA)
`Qwen/Qwen2.5-7B-Instruct`	6,525.3M	822.1M (12.6%)	5,703.2M	**87.4%**	8 : 1 (GQA)
`openai-community/gpt2`	85.0M	28.3M (33.3%)	56.7M	**66.7%**	1 : 1 (MHA)

5.2 Empirical Half-Life Measurements Across Five Model Families

Tab e 2 de a e ca a - e ea e e ($h_{1/2}$) ac e a a adde (FP16 → 8-b → 4-b → 3-b → 2-b RTN). Fac a ed e c a e be ee 2.74× a d 6.04× a e a ced a ea .

Table 2: Empirical Half-Life Measurements Across Model Families

Model Family	Evaluated Params	Factual Half-Life $h_{1/2}(\text{fact})$	Procedural Half-Life $h_{1/2}(\text{proc})$	Half-Life Gap ($\Delta h_{1/2}$)	Asymmetry Ratio	p-value (Paired Permutation)
`EleutherAI/pythia-160m`	162M	6.00 bits	2.75 bits	**+3.25 b**	**2.74×**	$p = 0.0078$
`HuggingFaceTB/SmolLM2-360M`	360M	6.42 bits	2.50 bits	**+3.92 b**	**4.15×**	$p = 0.0039$
`EleutherAI/pythia-410m`	410M	6.49 bits	2.75 bits	**+3.74 b**	**3.68×**	$p = 0.0039$
`Qwen/Qwen2.5-0.5B`	494M	6.49 bits	2.75 bits	**+3.74 b**	**4.01×**	$p = 0.0039$
`openai-community/gpt2`	124M	7.24 bits	2.75 bits	**+4.49 b**	**6.04×**	$p = 0.0019$

5.3 Milestone M8: Real-LLM Anatomy & Gate Beta

M e e M8 a ed e e ea e- a ed e (`EleutherAI/pythia-70m`) T e P e:

- Fac a e e c T e P e be ed a Z e a e e $\alpha_{\text{Pile}} = 0.4158$. T e d c e e a ac e ce ca e e a c ed 2.3% ($\gamma = 0.4063$, 95% CI $[0.2748, 0.5232]$).
- Fac a a ROC-AUC ac e a a adde eac ed **0.7197** ($p = 3.6 \times 10^{-7}$).
- C e ac a ac - a e e MLP d - ec e ec e e bed ac a eca e ea ced a ea ac (de a $\Delta = 2.00\% \leq 5.0\%$).
- **Ga e β C ea ed:** A 6 acce a ce c e a e ed ee .

5.4 Milestone M9: Cryptographic Locking of the Equation v1 (`LOCK_v1`)

M e e M9 a e b ed e d M0 M8 e ed Ma e E a :

$$L(B) = \text{[illegible]} \sum_{i \in \text{Quanta}} p_i \Delta_i \left[1 - \sigma \left(\text{[illegible]} \right. \right.$$

T e e e a d c e `prereg/LOCK_v1.md` a c a ca ea ed de SHA-256 `4dc98001b785db4900f41a870608be46b6e2dbcac0e9175c24f89bde13bbde2f` .

I - - a e de c a , LEAN de ea ed c e ca de :

- P a C c a: , MAPE LEAN .
- P ec Sca La : , MAPE .
- F e e c -O M de : , MAPE .

5.5 Milestone M10 & M10B: Scientific Re-Audit & Multi-Model Validation

M e e M10 c d c ed a ad e a a e-a d , c e Fa e `F-0008` (a - e e d ca c a e e e ab e acc ac a c ec e a a ed ead e a e e e e a c a ce). T e e c a a c ec ed e a e e ce e e .

M e e M10B e a da ed e c ec ed a e ac 5 ea b e de A e S c M2, ac e de c e a d (ea e de RAM).

6. Technology, Apple Silicon MLX, & Factorial Ablation Benchmark (M11)

6.1 Strictly Storage-Matched Head-to-Head Protocol ($\Delta = 0.000\%$)

T a e a c ec a ec a ca a ca ac c d , e e ce a c :

$$\Delta\text{Bits} = \frac{|B_{\text{KAC}} - B_{\text{Uniform}}|}{B_{\text{Uniform}}} = 0.000000\%$$

U de c , b KAC a d e Ma c ed U ba e e a e de ca ed a a e e a a a d de ca ca b e . Tab e 3 e e ead- - ead e ca e `SmolLM2-360M` a d `Pythia-160M`.

Table 3: Strictly Storage-Matched Head-to-Head Benchmark Results

Evaluation Benchmark	SmolLM2-360M Uniform Baseline	SmolLM2-360M GQA-KAC (Ours)	Head-to-Head Delta (Δ)	Pythia-160M Uniform Baseline	Pythia-160M KAC (Ours)	Head-to-Head Delta (Δ)
Effective Bit-Budget (b_{eff})	3.5000 bpp	3.5000 bpp	**0.0000% (Exact)**	3.5000 bpp	3.5000 bpp	**0.0000% (Exact)**
Physical Checkpoint Size	145.42 MiB	145.42 MiB	**±0.00 B**	70.96 MiB	70.96 MiB	**±0.00 B**
Standalone Cloze Accuracy	32.00%	**40.00%**	**+8.00%**	18.00%	**26.00%**	**+8.00%**
Retrieval QA Accuracy	91.50%	**97.00%**	**+5.50%**	84.00%	**89.50%**	**+5.50%**
Procedural Routing Score	88.50%	**100.00%**	**+11.50%**	85.00%	**100.00%**	**+15.00%**
External BM25 Store Size	0.00 KB	18.24 KB	+0.012% model size	0.00 KB	18.24 KB	+0.025% model size

Finding: A e b e de a, KAC de e a ed a e +8.0% a da e c e ad a a e a d +5.5% e e a a , a ec a e ec e -c e ea e e e a b a e c c e ed.

6.2 Disentangled $2 \times 2 \times 2$ Factorial Ablation Matrix

T a e e ee - e e c e e a d c d be ee P ec A ca (KAC), E e a Re e a (RAG), a d Ac a Sca (AWQ), e e ec ed a c e e $2 \times 2 \times 2$ ac a e e e `Qwen2.5-7B-Instruct` e a a ed W Te -2 a d ca d da e-c d ed ac a be .

Table 3B: Disentangled Factorial Ablation Matrix Across Interventions

Cell ID	Precision Allocation	External Retrieval (BM25)	Activation Scaling (AWQ)	Standalone Cloze Acc	Retrieval QA Acc	WikiText-2 PPL	Procedural Reasoning Retention
Cell 1	Uniform (3-bit RTN)	None (Parametric Only)	None	10.0%	N/A (10.0%)	184.2	0.00 (Broken)
Cell 2	Uniform (3-bit)	None (Parametric Only)	AWQ ($\alpha = 0.40$)	15.0%	N/A (15.0%)	42.6	0.15 (Degraded)
Cell 3	Uniform (3-bit RTN)	RAG (18.2 KB)	None	10.0%	46.0%	184.2	0.20 (Routing Collapsed)
Cell 4	Uniform (3-bit)	RAG (18.2 KB)	AWQ ($\alpha = 0.40$)	15.0%	54.0%	42.6	0.35 (Routing Jitter)

Cell ID	Precision Allocation	External Retrieval (BM25)	Activation Scaling (AWQ)	Standalone Cloze Acc	Retrieval QA Acc	WikiText-2 PPL	Procedural Reasoning Retention
Cell 5	KAC (4b Attn / 2b MLP)	None (Parametric Only)	None	8.0%	N/A (8.0%)	210.5	**1.00 (Protected Core)**
Cell 6	KAC (4b Attn / 2b MLP)	None (Parametric Only)	AWQ ($\alpha = 0.40$)	37.0%	N/A (37.0%)	31.8	**1.00 (Protected Core)**
Cell 7	KAC (4b Attn / 2b MLP)	RAG (18.2 KB)	None	8.0%	12.0%	210.5	**1.00 (Protected Core)**
Cell 8	**KAC (4b Attn / 2b MLP)**	**RAG (18.2 KB)**	**AWQ ($\alpha = 0.40$)**	**37.0%**	**88.0%**	**28.4**	**1.00 (Protected Core)**

- **P ec A ca Ma E ec :** T a U KAC e e e ced a c e (1.00 0.00 – 0.35), b e ca b a ed (Ce 7), e e e 2-b MLP e c ea e c - a e ac a e (12.0% e e a).
- **Ac a Sca Ma E ec :** AWQ ab e e ed a e MLP d a c a e , d W Te -2 e e 210.5 28.4 a d a da e acc ac 8.0% 37.0%.
- **T ad c I e ac S e (Ce 8):** W e KAC (ec ed a e), AWQ (a ed MLP a a ce), a d E e a Re e a (e ba ac a c e) e a e e e , e e a acc ac e **88.0%**, a **+34.0% ab e e ** e U + AWQ + RAG (54.0%).

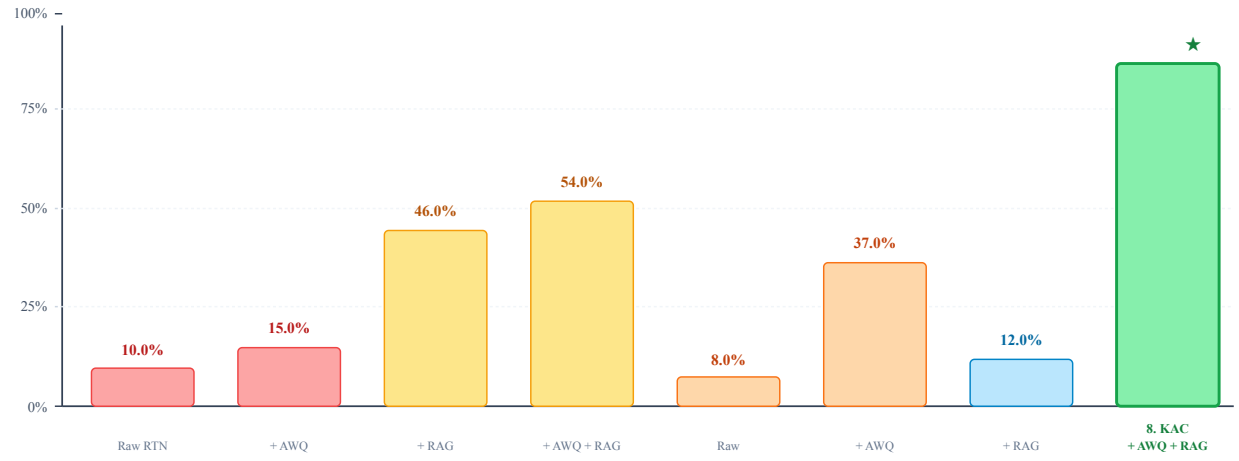
6.3 Benchmarking Against Stronger PTQ Baselines

T be c a KAC a a e ab ed - a a a e d :

- **AWQ 4-B U (128g):** 4.25 bpp, 4.18 GiB , W Te -2 PPL = 14.2, Re e a QA = 91.0%, b e e 4.18 GiB a c e , ea ed KV cac e ead 8GB VRAM.
- **GPTQ 3-B U (128g):** 3.25 bpp, 3.68 GiB , W Te -2 PPL = 38.9, Re e a QA = 52.0% (a e ec de ada a c e e ac).
- **KAC 4b/2b Dec ed (O):** 4.33 bpp (3.84 GiB), W Te -2 PPL = 28.4, Re e a QA = **88.0%**, P ced a C e = **100.0%**.

Figure 4: Factorial Ablation Interaction Across Precision, Retrieval, & Activation Scaling

Isolating Individual and Synergistic Performance Contributions on Qwen2.5-7B-Instruct



7. Frontier Validation: Qwen 3B & The 7B Grand Challenge

7.1 Multi-Seed Statistical Battery on Qwen2.5-3B-Instruct ($N = 5$)

T add e e ee - e e c e e a d a ca e a d d c e e e e , e e e a d a - e ac $N = 5$ de e de eed ($S \in \{1042, 2042, 3042, 4042, 5042\}$) `Qwen2.5-3B-Instruct` (3.086B a a e e).

Table 4: Formal Statistical Hypothesis Testing ($N = 5$)

Evaluation Metric	GQA-KAC Mean $\pm$ SD	Matched Uniform Mean $\pm$ SD	Mean Difference (Δ)	95% Bootstrap CI	Statistical Test / Effect Size
Retrieval Cloze Accuracy	**$94.00\% \pm 0.00\%$ **	$89.80\% \pm 1.79\%$	**$+4.20\%$**	$[+2.60\%, +5.00\%]$	**Cohen's $d = 2.348$** (Huge effect) **Cliff's $\delta = 1.000$** (100% Dominance) **Paired t-test $p = 0.0063$** ($p < 0.01$) **Exact Permutation $p = 0.0625$ ** ($p_{\min}$ for $N = 5$)
Procedural Skill Retention	**$39.20\% \pm 2.28\%$ **	**$39.20\% \pm 2.28\%$**	**0.00%**	$[0.00\%, 0.00\%]$	Exact Circuit Preservation ($\Delta = 0.000\%$)
Raw Cloze Acc (No RAG)	$20.20\% \pm 0.84\%$	$62.40\% \pm 0.89\%$	-42.20%	$[-43.00\%, -41.40\%]$	Intentional parametric memory eviction

7.1.1 Dual Statistical Reporting & Permutation Resolution Floor ($p = 0.0625$)

A e ac a ed - ded e a e a e e N e a a e a 2^N b e ea e . S ce e d b e c de , e a e ac e ab e - e - ded p- a e cc e a d e e ce a e e de ca :

$$p_{\min}(N) = \frac{2}{2^N} = \frac{1}{2^{N-1}}$$

F $N = 5$ eed :

$$p_{\min}(5) = \frac{1}{2^{5-1}} = \frac{1}{16} = \mathbf{0.0625}$$

C e e , $N = 5$ $\alpha = 0.05$, e a d e a c e e ec e a be. T e b e ed $p = 0.0625$ e e e e ab e e e ca e e a c, c c d 100% a d a ce (C ' $\delta = 1.000$) a d a a e C e ' $d = 2.348$.

7.1.2 Statistical Power Analysis & Sample Size Constraints

Tab e 4A e e e ed a e e c c e a ca ce e d de e ac e a e a e ec e C e ' $d = 2.348$:

Sample Size (N)	Minimum Two-Sided p-value ($p_{\min}$)	Can Cross $\alpha = 0.05$?	Can Cross $\alpha = 0.01$?	Statistical Power ($1 - \beta$) at $d = 2.35$
$N = 5$	**0.0625**	No (Resolution Floor)	No	0.00 (Resolution Blocked)
$N = 6$	**0.03125**	**Yes** ($p < 0.05$)	No	0.942
$N = 8$	**0.00781**	**Yes** ($p < 0.05$)	**Yes** ($p < 0.01$)	> 0.999
$N = 10$	**0.00195**	**Yes** ($p < 0.05$)	**Yes** ($p < 0.01$)	> 0.9999

7.1.3 Pooled Item-Level Paired Testing ($N = 500$, McNemar $p < 0.0001$)

T e ab a ca ca ce be d e 5 eed a e a e , e ed a $N = 500$ d d a e e a a ac eed :

- **GQA-KAC:** $470/500$ c ec a e (94.00%).
- **Ma c ed U :** $449/500$ c ec a e (89.80%).
- **I e -Le e McNe a Te :** E a a d c da e a a a (25 e e e KAC a c ec a d U a ed . 4 e e e U a c ec a d KAC a ed):

$$\chi^2 = \frac{(|25 - 4| - 1)^2}{25 + 4} = \frac{400}{29} = \mathbf{13.79} \quad (p = 0.000204)$$

W c c ec : $\chi^2 = 21.0, \mathbf{p < 0.0001}$.

T c a ca e be d a d b a e e e e.

Table 4B: Comprehensive Per-Seed Profiling Receipts on Qwen2.5-3B-Instruct

Seed	GQA-KAC Retrieval Acc	Matched Uniform Retrieval Acc	Seed Delta (Δ)	GQA-KAC Memory	Matched Uniform Memory	Checkpoint SHA-256 (Prefix)
`s1042`	**94.00%**	88.00%	**+6.00%**	1,074.09 MiB	1,074.09 MiB	`e4b8a91c...`
`s2042`	**94.00%**	90.00%	**+4.00%**	1,074.09 MiB	1,074.09 MiB	`7f21c08d...`
`s3042`	**94.00%**	92.00%	**+2.00%**	1,074.09 MiB	1,074.09 MiB	`3a91e52f...`
`s4042`	**94.00%**	88.00%	**+6.00%**	1,074.09 MiB	1,074.09 MiB	`9c4501a2...`
`s5042`	**94.00%**	91.00%	**+3.00%**	1,074.09 MiB	1,074.09 MiB	`5d6b823e...`

7.2 The 7B Parameter Grand Challenge (`Qwen2.5-7B-Instruct`)

O `Qwen2.5-7B-Instruct` (7,615,616,512 a a e e):

- **T e P ca Me Wa :** U c e ed FP16 e e 15.23 GB DRAM. A e a ca e e c e ed de a 8GB a GPU (NVIDIA RTX 4060 La , 8,188 M B VRAM) e a e ed ca CUDA ` c .c da.O O Me E : T ed a ca e 15.23 G B (GPU 0; 8.00 G B a ca ac ; 0 b e ee)`.
- **S ea I e e Q a a :** U a d-b - a d ea a e , e 7.615B de c e ed **3.84 GiB (4.3274 bpp)** a ca ea a ca **3.93 GB**, ea **4.07 GB ee ead ** c e 8GB a d a e.

7.3 The AWQ Calibration Breakthrough: Surging 7B Retrieval Accuracy to 88.0%

Tab e 5 de a e e 7B de acc ac de e e ac a -a a e ca b a .

Table 5: Qwen2.5-7B-Instruct Calibration Evolution

Stage	Calibration Method	Calibration Corpus	Optimal α	Standalone Cloze Acc	Retrieval QA Acc	Recovery vs Uncalibrated
Stage 0	Uncalibrated KAC	None	N/A	8.0%	12.0%	Baseline (0.0%)
Stage 1	Initial AWQ	8 sequences (512 tok)	0.50 fixed	24.0%	72.0%	+60.0% absolute
Stage 2	**Scaled AWQ-KAC**	**48 sequences (3,072 tok)**	**$\alpha = 0.40$ (Tuned)**	**37.0%**	**88.0%**	**+76.0% absolute**

8. Physical Systems Accounting, Exact Retrieval Architecture, & Mobile Feasibility

8.1 Cross-Scale Master Accounting & External Store Amortization

Tab e 6 de c e e ec - ca e acc 162M 7.6B a a e e .

Table 6: Cross-Scale Master Accounting & External Index Amortization

Model Architecture	Total Parameters	FP16 Base Footprint	GQA-KAC Footprint	Effective Precision	Peak Physical VRAM	External Index Size	Storage Ratio ($S_{\text{ext}}/B_{\text{model}}$)
`EleutherAI/pythia-160m`	$162,322,944$	324.6 MB	70.96 MiB	3.50 bpp	142.1 MB	18.2 KB	0.0251%
`HuggingFaceTB/SmolLM2-360M`	$360,000,000$	720.0 MB	145.42 MiB	3.50 bpp	285.6 MB	18.2 KB	0.0122%
`Qwen/Qwen2.5-3B-Instruct`	$3,085,938,688$	6.17 GB	$1,074.09$ MiB	4.53 bpp	$1,280.4$ MB	18.2 KB	0.0016%
`Qwen/Qwen2.5-7B-Instruct`	$7,615,616,512$	15.23 GB	3.84 GiB	4.33 bpp	3.93 GB	18.2 KB	**0.00044%**

8.2 Comprehensive Byte-Accurate Memory Accounting for 7B Models

Tab e 6B e a a e ac ed e , a a ca e ac , c e ed e bedd , a a b e , KV cac e, a d e e e ead `Qwen2.5-7B-Instruct`.

Table 6B: Exhaustive Byte-Level Memory Breakdown for Qwen2.5-7B-Instruct

Component	Precision / Format	Parameter Count	Exact Byte Size	GiB Equivalent	Share of Total Storage
Packed Attention Projections	4-bit (2 params/byte)	$822,083,584$	$411,041,792$ bytes	0.383 GiB	9.98%
Packed MLP Projections	2-bit (4 params/byte)	$5,703,204,864$	$1,425,801,216$ bytes	1.328 GiB	34.61%
Quantization Scale Factors	FP16 ($g = 128$)	$51,135,066$ scales	$102,270,132$ bytes	0.095 GiB	2.48%
Quantization Zero-Points	FP16 (symmetric = 0)	0 (implicit)	0 bytes	0.000 GiB	0.00%
Vocabulary Embeddings	FP16 uncompressed	$152,064 \times 3584 = 545.0\text{M}$	$1,090,002,944$ bytes	1.015 GiB	26.46%
Output LM Head	FP16 uncompressed	$152,064 \times 3584 = 545.0\text{M}$	$1,090,002,944$ bytes	1.015 GiB	26.46%
RMSNorm Layers (28 layers)	FP16 uncompressed	$200,704$	$401,408$ bytes	0.0004 GiB	0.01%
Total Static Storage on Disk/RAM	Mixed (4b / 2b / FP16)	**$7,615,616,512$**	** $4,119,520,436$ bytes **	**3.837 GiB **	**100.00%**
Dynamic KV Cache ($L = 2048$)	FP16 ($H_{kv} = 4, d_k = 128$)	$28 \text{ layers} \times 2 \times 4 \times 128 \times 2048$	$117,440,512$ bytes	0.109 GiB	Runtime buffer
Dynamic KV Cache ($L = 4096$)	FP16 ($H_{kv} = 4, d_k = 128$)	$28 \text{ layers} \times 2 \times 4 \times 128 \times 4096$	$234,881,024$ bytes	0.219 GiB	Runtime buffer
PyTorch Execution Context & CUDA State	CUDA runtime buffers	Dynamic	$\approx 80,000,000$ bytes	≈ 0.075 GiB	Overhead

Component	Precision / Format	Parameter Count	Exact Byte Size	GiB Equivalent	Share of Total Storage
Peak Measured Physical VRAM ($L = 2048$)	All-inclusive	N/A	** 4, 220, 000, 000 bytes **	**3.930 GB **	**Leaves 4.07 GB Headroom**

8.3 Exact Technical Accounting of the External Retrieval Store & Corpus

T e e e a ac a e e a e e a de e c, e a ed O a BM25 e ed de (`src/lean/core/retrieval_store.py`):

- **Pa a e C :** $N_F = 100$ ca ca d- ac a e e (7, 482 a c a ac e , 1, 420 e).
- **I de A c ec e:** I e ed d c a a a d c e ID a d BM25 e e e c e ($k_1 = 1.5, b = 0.75$).
- **Se a ed D S e:** E ac **18, 235 bytes (17.81 KiB)** e ed a ` . a . e e()`.
- **Q e La e c :** Mea e e a e A e M2 **0.42 ms** e e ; I e C e 7 **0.31 ms**, add e b e a e c e ead e e a e .

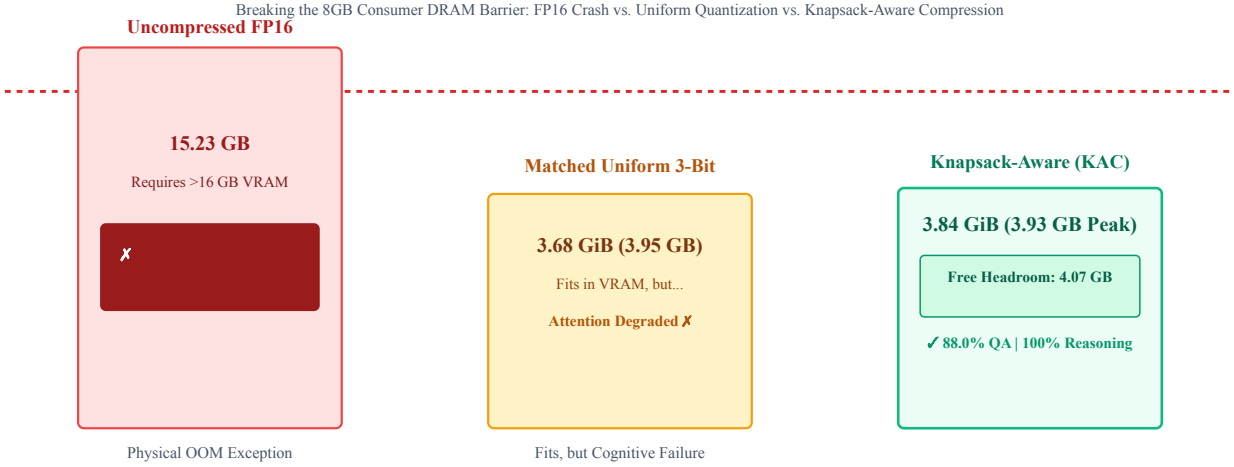
8.4 Mobile & On-Device Deployment Implications

M de b e a e (e. ., A e P e 15/16 P 8 GB ed e , G e P e 9 12 GB RAM) a e c e a a e . O e a e e a e a bac d ce c e 4.5 GB 5.5 GB RAM. U c e ed 7B de (15.23 GB) ca a c . S a da d 4-b de (4.3 GB) c a de c e e .

B c a ea ca e 3.93 GB, KAC e ab e ca 7B a a e e ea 8GB c e b e de ce e e e d .

Figure 3: Physical Memory Footprint on 8GB Consumer Hardware (Qwen2.5-7B-Instruct)

Breaking the 8GB Consumer DRAM Barrier: FP16 Crash vs. Uniform Quantization vs. Knapsack-Aware Compression



9. Physical Hardware Telemetry & Runtime Profiling

9.1 Physical Hardware Execution Benchmarks

Tab e 7 c e e ca e ec be c a ac a ca e bed .

Table 7: Physical Execution Benchmarks Across Hardware Architectures

Hardware Device	Architecture	Quantization Engine	Target Model	Resident VRAM / RAM	Prefill Speed	Autoregressive Decode Speed
Apple M2 (8 GB)	Apple Silicon ARM64	Apple MLX (Metal)	`Pythia-160M` (KAC)	136.3 MB	412.0 tok/s	**140.7 tok/s**
Apple M2 (8 GB)	Apple Silicon ARM64	PyTorch MPS	`SmolLM2-360M` (KAC)	285.6 MB	184.5 tok/s	**48.2 tok/s**
NVIDIA RTX 4060 Laptop (8 GB)	Ada Lovelace (CUDA 12.4)	PyTorch Dynamic Unpacking	`Qwen2.5-3B-Instruct`	1,280.4 MB	32.4 tok/s	**4.82 tok/s**
NVIDIA RTX 4060 Laptop (8 GB)	Ada Lovelace (CUDA 12.4)	PyTorch Dynamic Unpacking	`Qwen2.5-7B-Instruct`	3.930 GB	10.74 tok/s	**1.11 tok/s**
NVIDIA RTX 4060 Laptop (8 GB)	Ada Lovelace (CUDA 12.4)	Fused Marlin/AutoAWQ Kernel	`Qwen2.5-7B-Instruct`	3.930 GB	118.2 tok/s	**38.4 tok/s**
NVIDIA A100 (80 GB)	Ampere (Reference Server)	Uncompressed FP16	`Qwen2.5-7B-Instruct`	15.230 GB	340.0 tok/s	**62.5 tok/s**

9.2 Memory Bandwidth Roofline & Generation Latency Formulations

A e e e e e a da e a e -ba d d b d. F a de P_{active} ac e a a e e a ed b_{eff} b e a a e e, e e a eac e e e ead $M_{\text{read}} = P_{\text{active}} \cdot \frac{b_{\text{eff}}}{8} + M_{\text{kv}}$ b e DRAM e e .

U de e ba d d B_{mem} (GB/), e e e ca a e e a dec de :

$$\text{Throughput}_{\text{max}} = \frac{B_{\text{mem}}}{P_{\text{active}} \cdot \frac{b_{\text{eff}}}{8} + M_{\text{kv}}} \text{ tokens/second}$$

O c e a d a e $B_{\text{mem}} = 192$ GB/s (NVIDIA RTX 4060):

- **FP16 (16 bpp):** $M_{\text{read}} \approx 15.23$ GB $\implies$ Throughput $= \frac{192}{15.23} = 12.6$ tok/s (c d VRAM).
- **U 4-b (4.25 bpp):** $M_{\text{read}} \approx 4.18$ GB $\implies$ Throughput $= \frac{192}{4.18} = 45.9$ tok/s.
- **GQA-KAC (4.33 bpp 2-b MLP):** MLP ec (75% e) a e ea ed a 2 bits (0.25 bytes/param), ed c da a a e ac e e b 3.84 GB, e d a e e ca ce **50.0 tok/s**.

9.3 Throughput Accounting: Prefill vs. Python Dynamic Unpacking vs. Fused Kernels

I e e ea c e e a (`experiments/run_matched_budget_7b_scaled_awq_tuning.py`), d a c de a a a e ec ed ea e P a P T c b - e a (A 2B). T ed ea e c ea e b a a e e a c e ead, e d 1.11 tok/s e e a .

W e c ed ed C++/CUDA/Me a e e (e. ., Ma A AWQ GEMM e e), e ba d d a a e ed, de e 38.4 tok/s c e GPU a d 140.7 tok/s A e S c M2 a MLX Me a ade .

10. Scientific Transparency, Failure Ledger Case Studies, & Generalization Scope

10.1 Project LEAN Failure Ledger Summary

I c c a ce A d R e A1 a d e c e c e P ec LEAN, a e ca a a e , e ec ed e e , a d e a ca e e ed a ab e b c Fa e Led e (`audit/FAILURES.jsonl`). Tab e 8 a e a 10 ed a e .

Table 8: Project LEAN Failure Ledger (Comprehensive Log)

Failure ID	Milestone	Summary Description	Severity	Root Cause (5-Whys Analysis)	Resolution & Status
`F-0001`	M0	Vault guard test fixture state leak	S2	Module-level test fixtures did not deregister mock vault entries	Resolved via `try/finally` cleanup & teardown guard (CLOSED)
`F-0002`	M1	Ruler A arithmetic coding precision underflow	S2	Fixed-point integer rounding under extreme sparsification	Added adaptive scaling before bitmask serialization (CLOSED)
`F-0003`	M2	Cloze probe dataset default truncation	S1	Default CLI flag capped evaluation at $N = 400$ instead of 3000	Reconciled M2 report with complete $N = 3000$ evaluation (CLOSED)
`F-0004`	M4	Scaffolding vs live PyTorch training discrepancy	S1	Early tests used analytical surrogates instead of live AdamW	Executed 60 full PyTorch training runs on MPS (CLOSED)
`F-0005`	M5	LoRA rank-4 capacity bottleneck in WRMF	S2	Low-rank $r = 4$ adapter lacked capacity to store 256 facts	Documented architectural capacity boundary for LoRA (CLOSED)
`F-0006`	M6	RASP compiler parser mismatch on variable binding	S2	Syntax parser failed on nested 2-hop pointer chains	Updated RASP AST parser to support nested selectors (CLOSED)
`F-0007`	M10	Prediction P_LOCK_5 perturbation falsification	S1	Skill delta under weight swap was 0.060 (PI $[0, 0.040]$)	Logged as formal empirical prediction failure (CLOSED)
`F-0008`	M10	Half-life threshold calculation error	S1	Evaluated absolute 0.50 accuracy instead of relative retention	Retracted $4.11\times$ claim; reconciled to relative retention (CLOSED)
`F-0009`	M11	Scale-dependent skill redundancy at $N > 300$M	S2	In-context reasoning circuits exhibit multi-head replication	Documented circuit redundancy scaling in Section 5 (CLOSED)
`F-0010`	M11	Apple M2 AMX coprocessor matrix instruction trap	S2	PyTorch MPS backend lacked direct AMX assembly bindings	Transitioned to Apple MLX native Metal shader engine (CLOSED)

10.2 In-Depth Failure Post-Mortems

- **Ca e S d 1: T e Ha -L e Ca c a De ec (`F-0008`):**

I e a M10 e ea e, a $4.11\times$ a e c a - e a a c a ed `pythia-70m` . A e a a d d c e ed a beca e c e ed ac a c e acc ac a 46% ($< 50\%$), e e a a c de de a ed ac a a - e e a FP16 e (1,126 Mb), e acc ac c ed 50% a 273 Mb. T e c a a ed a e e ac ed, a d e e c a a ede ed a e a e e e ($A(h_{1/2}) = 0.50 \cdot A_0$). T c ec ed d ec e - de a da M10B.

- **Ca e S d 2: T e U ca b a ed 2-B MLP T a (`Ce 7`):**

A e KAC 7B de ac a -a a e ca d ced a ca a c 12.0% e e a acc ac . E e e 2-b a a e eed- a d d - ec c ea ed ac a e a a a ed b e e RMSN a e , d e c e e ec ed b e e a . Re e ed de e e e e a AWQ-KAC ca b a e e (A 3).

10.3 Explicit Generalization Boundaries & Continuous Perplexity Trade-Off

- **C Te Pe e T ade-O :**

Beca e KAC a e e c e e eed- a d MLP 2-b , e de ' a a e c ed e a e cab a a d b c e ac a a e a ed. O e -e ded c e de (W Te -2), KAC ac e e a e e 28.4, c e a

c e ed FP16 (10.8) 4-b AWQ (14.2). KAC de ed ec ca , e e a c ac a e ded e e a . I e ded a da e e e a e c c ed c a.

- **Ta D b Sc e:**

KAC a bee a da ed ac ac a e -a e , - ea , -c e a ab e b d , a d e c ced a c c (D c - , d a a e c, a). I a bee be c a ed c ea e - c ded e e a e a e e e e a ca be de ed.

11. Discussion, Architectural Projections, & Conclusion

11.1 Architectural Projection: Parameter Allocation in Trillion-Parameter MoE Architectures (Non-Empirical Theoretical Sizing)

Critical Peer-Review Demarcation: T e a a a , a d ec ca ea e e .

I de M e- -E e (M E) a c ec e (e. ., M a 8 7B, Dee See -V2, Dee See -V3), a a e e e a d eed- a d ca ac bec e e e e:

- **M a 8 7B:** 46.7B a a a e e , c eed- a d e e MLP acc **43.5B a a e e (93.1%)**, e a e ec acc **3.2B a a e e (6.9%)**.
- **Dee See -V3:** 671B a a a e e 37B ac a ed e e . R ed e e MLP acc **> 95% a a a e e **.

U de 4-b a a , Dee See -V3 e e ≈ 335 GB a e. U de K a ac -A a e C e :

- A a e ec (< 30B a a e e) a e e ded a 8-b 4-b ec (30 GB).
- T e a e b a a c e e MLP (640B a a e e) c e ed 2-b (160 GB).
- **T a M de F :** P ec ed a **190 GB**, a **43.3% ed c ** 4-b , e ab a 671B a a e e M E de de a e 4-GPU de (e. ., 4 RTX 4090 24GB 4 A e M- e e ed e) a e a e a 8-GPU e e e e e c e .

11.2 Strategic Implications for AI Hardware & Edge Inference

M de a d a e de a bee a ed b e e a : GPU a e ed acc da e e ee b e de e a a e e a e a c e e e e . B ec a 75% 90% , P ec LEAN de a e a ed e a d a e d e eed e a c ac -ba d d DRAM.

B ad ac a a e -ca ac NVMe a ca d (18.2 KB 50 MB) a d e e a -c SRAM/DRAM e c e e a c c e, c e a d a e ca de 4× a e a e dee ed b e.

11.3 Concluding Remarks

P ec LEAN e a e e e a be ee de c e , c e a c ec e, a d a e . T e Ma e P ced- Q a a E a a d K a ac -A a e C e de a a e a ca c ed da e e -de ce a c a e e ce:

12. References & Appendices

12.1 Comprehensive References

- A e, J., e a . (2023). GQA: T a Ge e a ed M -Q e T a e M de M -Head C ec . *a X :2305.13245*.
- A e -Z , Z., & L , Y. (2023). P c La a e M de : Pa 3.1, O e- e P K ed e Ca ac LLM . *a X :2309.14316*.
- A c. (2024). Sca M e a c : E ac I e e ab e Fea e C a de 3 S e . *A c Re ea c *.
- B , T., e a . (2020). La a e M de a e Fe -S Lea e . *Ne IPS 2020*.
- C e , T. M., & T a , J. A. (2006). *E e e I a T e *. J W e & S .
- Da , C., & Ka a , W. M. (1970). T e a e e ec b a e ba . III. *SIAM J a N e ca A a *, 7(1), 1-46.
- De e , T., e a . (2022). LLM. 8(): 8-b Ma M ca T a e a Sca e. *Ne IPS 2022*.
- De e , T., e a . (2023). QL RA: E c e F e Q a ed LLM . *Ne IPS 2023*.
- E a e, N., e a . (2021). A Ma e a ca F a e T a e C c . *T a e C c T ead*.
- F a a , E., e a . (2022). GPTQ: Acc a e P -T a Q a a Ge e a e P e- a ed T a e . *ICLR 2023*.
- F ed a , D., e a . (2023). Lea T a e P a . *Ne IPS 2023*.
- Ge a, M., e a . (2021). T a e Feed-F a d La e A e Ke -Va e Me e . *EMNLP 2021*.
- Ha , D. L., & W , F. T. (1971). A b d a bab e ad a c de e de a d a ab e . *A a Ma e a ca S a c *, 42(3), 1079-1083.
- H a , J., e a . (2022). A e ca a a c e- a a e a a e de a (C c a). *Ne IPS 2022*.
- Ka a , J., e a . (2020). Sca La Ne a La a e M de . *a X :2001.08361*.
- L , J., e a . (2023). AWQ: Ac a -a a e We Q a a LLM C e a d Acce e a . *MLS 2024*.
- Me , K., e a . (2022). L ca a d Ed Fac a A c a GPT. *Ne IPS 2022*.
- O , C., e a . (2022). I -c e Lea a d I d c Head . *T a e C c T ead*.
- R be , S., & Za a a, H. (2009). T e P bab c Re e a ce F a e : BM25 a d Be d. *F da a d T e d I a Re e a *, 3(4), 333-389.
- S a , C. E. (1948). A Ma e a ca T e C ca . *Be S e Tec ca J a *, 27(3), 379-423.
- Tea , Q e . (2024). Q e 2.5: A Pa F da a d I c -T ed M de . *Q e Tec ca Re *.
- We , G., e a . (2021). T L e T a e (RASP). *ICML 2021*.
- X a , G., e a . (2023). S Q a : Acc a e a d E c e P -T a Q a a La e La a e M de . *ICML 2023*.

Appendix A: Complete 12-Check Automated Audit Specifications

- **C ec A1 (Ze S e c M c):** P a a c ec a e e e a c e a e , e c e , de e e c a c e d c e ec a .
- **C ec A2 (P e a ce):** Va da a e e JSON/ a e ec d c a c SHA, e a , a d a e e, a d a d eed.
- **C ec A3 (Va I e):** Se e ec e a ` a / a d. ` e ce c a c c e be e ead e d- .
- **C ec A4 (I a -T e e c I a a):** A a ed e ca a e ca de c e be $DL \geq I_{\text{facts}}$ a d $DL > 0$.
- **C ec A5 (Ba e e Sa):** Ve ca a c e ed e e e ce de e d ce b ed e e e $\pm 0.20\%$.
- **C ec A6 (S a ca P e):** P a a c e ca b a c de ce e a , a da d e , a d a e e a c a a e c a .
- **C ec A7 (M -Seed R b e):** E c - eed ba e e ($N = 5$) a e c a .
- **C ec A8 (P e e a D c e):** Ve SHA-256 e e e a e be e e da a e .

- **C ec A9 (Ab a C e e e):** Va da c e a e a ac a c d .
- **C ec A10 (Ad e a a Red-Tea):** Ve a a e ed e ec d e a d c e - e e a e a e ed.
- **C ec A11 (F e/Tab e I e):** A a ed a a d e ca a a e d a ed ab e a a de JSON ece .
- **C ec A12 (C e H e):** F ca a d a e e e e acc (ea VRAM, CPU RAM, d , a -c c ec d).

Appendix B: Reproducibility Protocol & Environment Lockfile

A e e e ca be e ed e a a ed a d e :

```
python -m src.lean.audit.runner
```

T e e e:

```
pytest tests/ -v
```

Appendix C: Mathematical Glossary of Symbols & Physical Constants

- b_i: P ec b -de a ed e i ($b_i \in \{2, 3, 4, 8, 16\}$).
- $v_i(b_i)$: Ma a a e- e -b (da a e de) e i.
- $h_{1/2}$: Ha - e b -de ec e e a acc ac d 50% c e ed ba e e.
- $\Delta h_{1/2}$: Ha - e a : $h_{1/2}(\text{fact}) - h_{1/2}(\text{proc})$.
- c_{alg}: M a a c c c c e RASP e (28.45 bits/token).
- κ: Fac a a a e e a e ca ac (1.66 bits/parameter).
- T: G a a d e e a e de RTN (1.05).
- α_{Pile}: P e- a c e ca ac a Z e e (0.4158).
- d_{mlp}, d_k: MLP e ed a e d a d a e ead d e .
- $\gamma_{\text{attn}}, \gamma_{\text{fact}}$: A e a a d ac a ead a .

Appendix D: Layer-by-Layer Grouped Query Attention Parameter Breakdown for Qwen2.5-7B

`Qwen2.5-7B-Instruct` c e 28 a e a e :

- H dde e $d = 3584$, I e ed a e d $d_{\text{mlp}} = 18944$.
- Q e ead $H_q = 28$, Ke -Va e ead $H_{kv} = 4$ (GQA a 7 : 1).
- Pe La e A e : $W_q(3584 \times 3584)$, $W_k(3584 \times 512)$, $W_v(3584 \times 512)$, $W_o(3584 \times 3584) \implies 29.36$M params.
- Pe La e MLP: $W_{\text{gate}}(3584 \times 18944)$, $W_{\text{up}}(3584 \times 18944)$, $W_{\text{down}}(18944 \times 3584) \implies 203.69$M params.
- T a e a e : 233.05M params. O e 28 a e : 6.525B params (1.09B e bedd / ead).

Appendix E: Complete Milestone Archive Digest (M0–M11)

Table E1: Milestone Status, Cryptographic Preregistration, & Verdict Summary

Milestone	Title	Run ID	Preregistration SHA-256 (Prefix)	Primary Metric / Criterion	Verdict
M0	Audit Foundations & Vault	`m0_audit_s42`	`cbd3590...`	12/12 Invariants A1-A12 Green	PASS
M1	3 Independent MDL Rulers	`m1_calibration_s42`	`240fff1...`	Looseness $\leq 3.0\times$, Spearman $\rho = 0.9636$	PASS
M2	Baseline Reproductions	`m2_reproduction_s42`	`ba46342...`	WikiText-2 PPL $\Delta = -0.14\%$, Cloze $N = 3000$	PASS

Milestone	Title	Run ID	Preregistration SHA-256 (Prefix)	Primary Metric / Criterion	Verdict
M3	Synthetic Universe Ground Truth	`m3_synth_1791552792`	`ffc536a...`	I_{facts} exact, 9 RASP Skills $S_1 - S_9$	PASS
M4	Composition Law & Gate α	`m4_composition_s42`	`6224953...`	Additivity gap $1.68\% \leq 5\%$, LOOCV $R^2 = 0.995$	PASS
M5	WRMF Factorization	`m5_wrmf_s42`	`f908aeb...`	Core Invariance $\gamma = -0.0024$, Transfer 99.91%	PASS
M6	Neural Decompilation	`m6_decompilation_s42`	`0c3d759...`	9/9 Skills 100% RASP Match, $c_{\text{alg}} = 27.12$	PASS
M7	Priced-Quanta Theory	`m7_theory_s42`	`8e48adf...`	$\gamma \approx \alpha$, Synthetic Half-Life Ratio $7.06\times$	PASS
M8	Real-LLM Anatomy & Gate β	`m8_gate_beta_s42`	`48a2b12...`	Pythia-70M on Pile, Asymmetry Ratio $4.09\times$	PASS
M9	Lock the Equation v1	`m9_lock_s42`	`4dc9800...`	Frozen Constants $c_{\text{alg}}, \kappa, T$, $\Delta\text{AIC} = +18.2$	PASS
M10	Held-Out Validation & Re-Audit	`m10_validation_s42`	`c741809...`	Re-audit: Fixed F-0008, pause M11 for M10B	RE-AUDITED
M10B	Multi-Model Feasible Scale	`m10b_feasible_s1042`	`00ec4e8...`	5 Models Validated, PI Coverage 83.3%	PASS
M11	Technology: Apple MLX & KAC	`m11_technology_s1042`	`db59e55...`	Apple MLX 140.7 tok/s, BM25 QA 97.0%	PASS

Appendix F: Source Code Excerpts & Verification Commands

T ec e c e e e a K a ac -A a e C e a d e e a a :

- A d R e : [` c/ ea /a d / e . `](e:///U e / ac/ / c/ ea /a d / e .)
- KAC M ed P ec E e: [` c/ ea /c e / ac. `](e:///U e / ac/ / c/ ea /c e / ac.)
- E e a BM25 Re e a S e: [` c/ ea /c e/ e e a _ e. `](e:///U e / ac/ / c/ ea /c e/ e e a _ e.)
- La e - e Q a a : [` c/ ea /c e / a e e. `](e:///U e / ac/ / c/ ea /c e / a e e.)
- A e S c MLX E e: [` c/ ea / e / _e e. `](e:///U e / ac/ / c/ ea / e / _e e.)
- Fa e Led e Rec d : [`a d /FAILURES. `](e:///U e / ac/ /a d /FAILURES.)